Highlights

TRACER-Med: Trustworthy Deep Learning for Differentially Private, Equitable, and Uncertainty-Calibrated Clinical Decision Support

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- Formulates clinical Privacy-Fairness-Uncertainty Trilemma under DP.
- Derives Riemannian Dirichlet-Fisher DP-SGD from information geometry.
- Proves strict per-sample record privacy for class prototype transport.
- Establishes finite-sample sub-exponential PAC conformal bounds.
- Dual-channel Dirichlet XAI isolates disease risk from clinical doubt.

TRACER-Med: Trustworthy Deep Learning for Differentially Private, Equitable, and Uncertainty-Calibrated Clinical Decision Support

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Abstract

Deploying deep neural networks in clinical decision support requires simultaneously satisfying strict patient privacy statutes, demographic error-rate equity, calibrated uncertainty quantification, and clinically actionable explainability. However, integrating standard Differentially Private Stochastic Gradient Descent (DP-SGD) with evidential learning induces a Privacy–Fairness–Uncertainty Trilemma: uniform gradient clipping disproportionately attenuates minority representations, privacy perturbation destabilizes Dirichlet evidence accumulation, and private conformal calibration risks set vacuity on small cohorts. To resolve these tensions, we introduce **TRACER-Med**, an organically unified trustworthy deep learning architecture for biomedical decision support. TRACER-Med integrates: (1) an evidential Dirichlet network decomposing predictive uncertainty into aleatoric variation and epistemic vacuity; (2) Riemannian Evidential-Fisher Subgroup-Calibrated DP-SGD (REF-DP), deriving adaptive cohort clipping thresholds from the Dirichlet Fisher metric to preserve minority representations under Rényi Differential Privacy; (3) Privatized Class-Conditional Prototypical Anchor Transport (PC-PAT), contracting the 2-Wasserstein demographic distance to learned diagnostic prototypes with strict per-sample gradient independence; (4) Differentially Private Group-Conditional Conformal Prediction (DP-GCCP), implementing an Empirical-Bayes shrinkage estimator over Laplace-perturbed histograms with bounded replacement ℓ_1 sensitivity $\Delta_1 = 2$ and sub-exponential PAC concentration bounds; and (5) Dual-Channel Dirichlet Attribution (DCDA), disentangling diagnostic disease risk from epistemic clinical doubt relative to standardized physiological reference baselines. Validated across Statlog Heart Disease ($N = 270$), Pima Indians Diabetes ($N = 768$), and a simulated acute inpatient EHR cohort ($N = 10{,}000$) against ERM, DP-SGD, a DPSGD-F-style group-clipping baseline, Fair-ERM, evidential learning, and DP-EDL, the prototype alignment reduces demographic parity disparity on Statlog Heart from 0.285–0.324 for private baselines to 0.094 and equalized-odds disparity from 0.137–0.168 to 0.075, with similar gains on Pima (Δ_{SP} 0.235 versus 0.325–0.377). It attains the lowest empirical membership-inference AUC among private models on all three cohorts (0.519, 0.511, 0.502) and, on the large cohort, lower calibration error than the private baselines (ECE 0.080 versus 0.146–0.219). Under a rigorous Poisson-subsampled RDP accountant, the privacy budgets are large ($\epsilon = 27.7$, 17.5, and 7.3 at $\delta = 10^{-5}$), and the private conformal layer yields uninformative prediction

sets on the two small cohorts, a limitation we analyze together with ablations that show no measurable benefit of subgroup-calibrated clipping beyond prototype alignment in these experiments. TRACER-Med couples subgroup-aware private optimization, class-conditional fairness alignment, private conformal calibration, and uncertainty-aware attribution under a single sensitivity analysis, and is accompanied by formal privacy, transport, coverage, and attribution guarantees.

Keywords: Trustworthy Machine Learning, Differential Privacy, Algorithmic Fairness, Evidential Deep Learning, Conformal Prediction, Clinical Decision Support

1. Introduction

1.1. The Translational Frontier of Clinical Artificial Intelligence

The integration of deep learning into biomedical practice marks a transformative paradigm shift for clinical decision support systems (CDSS) [1, 2, 3]. Across intensive care units (ICUs), emergency departments, cardiovascular wards, and ambulatory endocrinology clinics, modern hospital networks continuously aggregate multidimensional patient streams—ranging from high-frequency physiological telemetry and longitudinal electronic health records (EHRs) to comprehensive laboratory biomarker panels. Neural architectures trained on these heterogeneous data streams offer unprecedented potential for early disease detection, dynamic risk stratification, and patient triage prior to overt physiological collapse [2, 4].

However, despite extraordinary retrospective benchmarking achievements, an acute *translational chasm* separates in-silico model development from safe, prospective bedside deployment. Unlike consumer machine learning applications where marginal inaccuracies incur negligible risk, clinical decision-making is high-stakes, irreversible, and governed by strict ethical oaths and biomedical standards. When a clinical model is deployed at the bedside, its predictions directly inform invasive diagnostics, therapeutic interventions, intensive care admissions, or discharge decisions. In this unforgiving environment, deep neural networks frequently exhibit silent, catastrophic vulnerabilities: they memorize private genomic and biometric records; they perpetuate systemic diagnostic disparities against underrepresented patient demographics; they generate dangerously overconfident hallucinations on ambiguous or atypical presentations; and they produce ungrounded, opaque saliency maps that fail to provide actionable pathophysiology.

1.2. A Motivating Bedside Clinical Dilemma

To appreciate why conventional deep learning systems fail in acute clinical settings, consider a high-stakes emergency department encounter. An elderly female patient presents with vague, non-classical clinical indicators: intermittent epigastric discomfort, nausea, mild dyspnea, and atypical fatigue, accompanied by borderline glycemic elevation. In female

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and elderly cohorts, ischemic heart disease frequently manifests through these non-anginal presentations rather than classic substernal chest pressure radiating to the left arm [5, 6].

Suppose an emergency clinician consults an institutional CDSS to evaluate the probability of an acute coronary syndrome (ACS). If this system is built using contemporary modular machine learning techniques, it threatens the patient's safety through a cascade of compounding failures:

1. **Privacy-Induced Disparity Penalty:** To comply with patient privacy statutes, the model was trained using standard Differentially Private Stochastic Gradient Descent (DP-SGD) [7]. Because elderly female cardiac cases constitute a numerical minority with high physiological variance, their per-sample gradients were disproportionately truncated by uniform Euclidean clipping thresholds. The isotropic noise injected during private optimization destroyed the subtle gradient signals defining female atypical presentations, inducing a severe false-negative bias [8, 9].
2. **Softmax Overconfidence and Ignorance Concealment:** Parameterized by standard softmax normalization over uncalibrated logits, the model outputs an assertive point prediction: "86% probability of non-cardiac etiology." Because softmax transforms arbitrarily large negative logits into normalized probabilities summing to one, the network is fundamentally incapable of signaling epistemic ignorance (i.e., "I have never seen an atypical presentation of this specific biomarker configuration under differential privacy") [10, 11]. The physician receives false reassurance.
3. **Absence of Finite-Sample Conformal Safety Sets:** The clinician is given a single, uncalibrated probability without distribution-free statistical coverage guarantees. Without group-conditional conformal prediction sets, there is no mathematical guarantee that a prediction set (e.g., $\{\text{Non-Ischemic}, \text{NSTEMI}\}$) will contain the ground truth with $1 - \alpha_{\text{conf}}$ confidence specifically for this female demographic cohort [12, 13].
4. **Conflated and Unactionable Attribution:** An ad-hoc single-channel saliency heatmap points to the patient's elevated glucose level. The physician is left in the dark: is the model predicting non-cardiac etiology because glucose indicates metabolic distress, or is the model highly uncertain because troponin and electrocardiogram readings are ambiguous? Standard explainability methods conflate diagnostic risk factors with drivers of epistemic vacuity, preventing the clinician from determining whether to order serial troponin assays or discharge the patient [14, 15].

1.3. The Regulatory and Institutional Governance Imperative

In response to these compounding risks, international health authorities and legislative bodies have established rigorous regulatory mandates that render single-objective, black-box clinical algorithms obsolete:

- **FDA Good Machine Learning Practice (GMLP):** Jointly developed by the US Food and Drug Administration (FDA), Health Canada, and the UK Medicines and Healthcare products Regulatory Agency (MHRA), GMLP Principle 7 mandates that medical AI models must demonstrate performance stability, fairness, and safety across relevant patient sub-populations throughout their total product lifecycle [16].

- **FDA Predetermined Change Control Plans (PCCP):** The FDA's finalized guidance on PCCP for AI/ML-enabled device software functions requires device sponsors to specify robust post-market monitoring protocols ensuring that algorithmic updates do not introduce demographic degradation, calibration drift, or privacy vulnerabilities [17].

- **European Union Artificial Intelligence Act (EU AI Act):** Under Regulation (EU) 2024/1689, AI systems intended for medical diagnosis and clinical risk prediction are classified as *High-Risk AI Systems* (Annex III) [18]. Articles 9–15 enforce binding legal obligations covering continuous risk management (Art. 9), rigorous data governance and bias auditing (Art. 10), technical documentation and record-keeping (Arts. 11–12), transparent clinician-facing explainability (Art. 13), meaningful human oversight (Art. 14), and high standards of accuracy, robustness, and cybersecurity (Art. 15).

- **Clinical Reporting Standards (TRIPOD+AI and FUTURE-AI):** Methodological consensus frameworks such as TRIPOD+AI [19] and FUTURE-AI [20] emphasize that biomedical models cannot be evaluated solely on aggregate AUC-ROC. They mandate rigorous auditing of subgroup-specific error rates, probabilistic calibration, and privacy-preserving data handling.

1.4. The Privacy–Fairness–Uncertainty Trilemma

To satisfy these regulatory and clinical demands, contemporary clinical data science typically adopts a fragmented, modular approach: engineers train an architecture with DP-SGD [7] to satisfy HIPAA/GDPR constraints; apply post-hoc demographic parity or equalized odds thresholding [21] to mitigate bias; apply temperature scaling or MC-dropout [22] for uncertainty estimation; and compute SHAP or Integrated Gradients [23, 24] for post-hoc interpretability.

We demonstrate that this modular assembly fundamentally breaks down because privacy, fairness, uncertainty, and explainability are deeply mathematically and operationally entangled. We formalize this interaction as the **Privacy–Fairness–Uncertainty Trilemma**:

- **Trade-off 1 (Differential Privacy Disproportionately Truncates Minority Gradients):** In standard DP-SGD, per-sample gradients are clipped to a uniform Euclidean threshold C, followed by isotropic Gaussian noise addition. Because underrepresented clinical cohorts exhibit distinct physiological presentations and smaller sample counts $N_a \ll N$, their loss gradients reside in high-variance, high-norm regimes. Uniform Euclidean clipping disproportionately truncates minority gradient updates, while the uniform noise floor degrades their effective Signal-to-Noise Ratio (SNR). This creates an algorithmic "equity penalty," widening demographic error disparities and exacerbating under-diagnosis [8, 9, 25, 26].

- **Trade-off 2 (Privacy Noise Destabilizes Probability Calibration and Dirichlet Evidence):** In evidential deep learning frameworks, Dirichlet concentration parameters $\boldsymbol{\alpha}$ represent collected clinical evidence. The gradient of evidential losses scales inversely with accumulated evidence. When DP-SGD is applied naively, uniform clipping

limits the rate at which rare disease presentations can accumulate evidence, while isotropic Gaussian perturbation corrupts the Dirichlet concentration parameters. This evidence suppression distorts probability calibration, inflating Expected Calibration Error (ECE) and producing uncalibrated, overconfident predictions in high-risk clinical regimes [27, 10].

- **Trade-off 3 (Subgroup Conformal Safety vs Prediction Set Cardinality):** Split conformal prediction provides rigorous, distribution-free marginal coverage guarantees on the aggregate population [13, 12, 28]. However, marginal guarantees permit severe undercoverage on protected demographic cohorts. Enforcing group-conditional coverage under differential privacy requires private quantile computation over small cohort calibration sets n_a. To guarantee Probably Approximately Correct (PAC) coverage with high confidence $1 - \beta_{\text{conf}}$, the quantile estimate must incorporate finite-sample and privacy noise concentration buffers $\tau_a \propto \frac{1}{\sqrt{n_a}} + \frac{1}{n_a \epsilon_{\text{conf}}}$. For small minority cohorts, this buffer inflates the non-conformity threshold, creating an acute tension between subgroup coverage equity and prediction set efficiency (set cardinality $|\mathcal{C}(\mathbf{x})|$) [29].

Because these trade-offs arise from fundamental mathematical tensions among gradient norms, privacy noise geometry, and finite-sample concentration, they cannot be resolved through post-hoc calibration or independent pipeline stages. Resolving the Trilemma requires an organically unified, first-principles architecture.

1.5. Our Contributions

To resolve the Privacy–Fairness–Uncertainty Trilemma within a cohesive architectural paradigm, we propose **TRACER-Med** (**TR**ustworthy, **A**ligned, **C**onformalized, **E**vidential, and Differentially P**R**ivate Deep Learning for **Med**icine). TRACER-Med synthesizes information-geometric differential privacy, class-conditional prototype alignment, distribution-free private conformal calibration, and dual-channel evidential attribution. Our principal contributions are:

1. **Formulation of the Clinical Trilemma:** We provide a formal characterization of the Privacy–Fairness–Uncertainty Trilemma in biomedical deep learning, proving why naive modular combinations of DP-SGD, fairness post-processing, and evidential learning induce compounding trade-offs on imbalanced clinical cohorts.
2. **Riemannian Evidential-Fisher Subgroup-Calibrated DP-SGD (REF-DP):** We reformulate evidential optimization through the lens of information geometry on the Dirichlet statistical manifold. Deriving the exact closed-form inverse Fisher Information Metric $\mathcal{G}^{-1}(\boldsymbol{\alpha})$ via the Sherman–Morrison formula, we compute Riemannian Dirichlet natural gradients and derive adaptive subgroup clipping thresholds C_a^* proportional to trace curvature $\text{Tr}(\mathcal{G}(\bar{\boldsymbol{\alpha}}_a))$ and external epidemiological priors p_a. In Theorem 3, we prove that bounding per-sample gradients by C_a^* and scaling noise to worst-case sensitivity $C_{\max} = \max_a C_a^*$ provides strict Rényi Differential Privacy (RDP) guarantees while eliminating the minority gradient truncation penalty.

3. **Privatized Class-Conditional Prototypical Anchor Transport (PC-PAT):** Rather than enforcing marginal demographic parity which distorts disease base rates, we formulate class-conditional alignment toward invariant, learnable diagnostic prototypes $\bar{\boldsymbol{\mu}}_y$. In Proposition 2, we prove that minimizing our prototypical anchor loss directly upper-bounds the demographic 2-Wasserstein transport distance $\mathcal{W}_2(\mathbb{P}_{0,y}, \mathbb{P}_{1,y}) \to 0$. In Theorem 3, we prove that because prototypes are learnable model parameters updated via gradient descent, the loss preserves strict per-sample record independence, eliminating cross-sample coupling and securing differential privacy sensitivity.
4. **PAC-Guaranteed Differentially Private Group-Conditional Conformal Prediction (DP-GCCP):** We establish a private conformal calibration mechanism over disjoint score histograms with bounded replacement ℓ_1 sensitivity $\Delta_1 = 2$. In Theorem 4, we prove that integrating empirical CDF concentration with sub-exponential Laplace tail bounds guarantees $(\epsilon_{\text{conf}}, 0)$-DP and $(\beta_{\text{conf}}, 1 - \alpha_{\text{conf}})$-PAC coverage equity across demographic cohorts. We formalize the trade-off between pure PAC thresholds and operational shrinkage for small clinical cohorts.
5. **Dual-Channel Dirichlet Attribution (DCDA):** Exploiting Dirichlet information geometry, we formulate a dual-channel path-integrated gradient mechanism satisfying axiomatic Completeness, Implementation Invariance, and Linearity (Proposition 5). Channel A isolates input features driving diagnostic posterior risk, while Channel B isolates features driving Epistemic Mutual Information relative to standardized physiological baselines, providing clinicians with actionable diagnostic transparency.
6. **Multi-Cohort Empirical Validation and Governance Mapping:** We validate TRACER-Med across three clinical cohorts: Statlog Heart Disease ($N = 270, D = 12$), Pima Indians Diabetes ($N = 768, D = 8$), and an acute inpatient EHR cohort ($N = 10{,}000, D = 15$). We report where TRACER-Med improves on private baselines (demographic parity and Equalized Odds on the two clinical cohorts, membership-inference resistance, and calibration on the large cohort) and where it does not (conformal set informativeness on small cohorts, calibration on Statlog Heart, and any benefit of subgroup-calibrated clipping over prototype alignment alone). Finally, we map our framework directly to FDA GMLP, FDA PCCP, and EU AI Act regulatory requirements.

1.6. Structure of the Article

The remainder of this article is organized as follows. Section 2 reviews related work in differential privacy, algorithmic fairness, evidential uncertainty quantification, and explainability. Section 3 formalizes the biomedical setting, notation, and threat model. Section 4 details the mathematical architecture of TRACER-Med, presenting REF-DP, PC-PAT, DP-GCCP, and DCDA alongside theoretical proofs. Section 5 describes the experimental benchmarks, baselines, and evaluation protocols. Section 6 presents empirical results, ablation studies, and qualitative clinical case studies. Section 7 discusses clinical governance, regulatory alignment, and limitations, and Section 8 concludes the article.

2. Related Work

2.1. Differential Privacy in Biomedical Machine Learning

Differential Privacy (DP) [30] provides a rigorous mathematical framework bounding the privacy risk of individual training records. In deep learning, DP-SGD [7] limits per-sample gradient ℓ_2 norms to threshold C, injects Gaussian perturbation $\mathcal{N}(0, \sigma_{\text{DP}}^2 C^2 \mathbf{I})$, and accumulates privacy loss via Rényi Differential Privacy (RDP) [31, 32] or Gaussian Differential Privacy (GDP) [33].

However, Bagdasaryan et al. [8] and Suriyakumar et al. [9] demonstrated that DP-SGD acts as an equity penalty: uniform gradient clipping severely truncates underrepresented patient cohorts, widening demographic error disparities. Subsequent investigations explored fair DP in generic classification: Xu et al. [25] proposed DPSGD-F, adapting group clipping thresholds to balance accuracy loss; Tran et al. [26] formulated Lagrangian dual optimization for fair private empirical risk minimization; and Esipova et al. [34] analyzed gradient misalignment driving disparate impact under DP. Furthermore, Papernot and Steinke [35] formalized private hyperparameter selection to prevent unaccounted privacy leakage. However, these methods focus primarily on point-prediction parity metrics. They do not address evidential uncertainty decomposition, distribution-free conformal calibration, or dual-channel explanations. TRACER-Med bridges this divide by grounding subgroup clipping in Dirichlet information geometry, unified with class-conditional optimal transport and private conformal calibration.

2.2. Algorithmic Fairness in Clinical Healthcare

Biomedical algorithmic bias is well-documented. Obermeyer et al. [36] demonstrated that commercial healthcare algorithms systematically prioritized healthier White patients over sicker Black patients for high-risk care management programs. Seyyed-Kalantari et al. [6] revealed that chest radiograph classifiers exhibited under-diagnosis bias in female and underserved populations. Typical fairness interventions comprise pre-processing, in-processing (e.g., fair representation learning [37] and representation debiasing via Sliced Wasserstein Distance [38, 39, 40]), and post-processing (e.g., equalized odds thresholding [21]). However, existing debiasing techniques often target marginal demographic parity, which is clinically counterproductive under unequal base rates [41, 42, 43]. Reporting guidelines such as TRIPOD+AI [19] and consensus frameworks like FUTURE-AI [20] emphasize that algorithmic fairness and representativeness must be evaluated through condition-specific error-rate parity rather than crude demographic parity. TRACER-Med focuses on class-conditional latent alignment targeting Equalized Odds directly within private optimization.

2.3. Evidential Uncertainty Quantification and Conformal Prediction

Standard deep neural networks with softmax parameterizations tend to output over-confident probabilities on ambiguous clinical profiles [4, 44, 22]. Evidential Deep Learning (EDL) [10] parameterizes a Dirichlet distribution over categorical probability vectors, allowing analytical decomposition into aleatoric entropy and epistemic vacuity. While recent studies have examined theoretical nuances of loss-minimized evidential learning [11, 45], evidential

parameterizations offer closed-form analytical properties suitable for path-integrated gradient explanations.

Conformal Prediction (CP) [13, 12, 28] constructs distribution-free, finite-sample prediction sets $\mathcal{C}(X)$ satisfying marginal coverage guarantee $\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha_{\text{conf}}$. To enforce equity across demographic subgroups, group-conditional and Mondrian conformal prediction [46, 47, 48, 49] compute cohort-specific non-conformity quantiles. Extending conformal prediction to differentially private settings requires private quantile mechanisms [29] to avoid membership leakage while preserving bounded set size efficiency.

2.4. Explainable AI and Attribution Robustness

Attribution methods such as Integrated Gradients (IG) [23] and SHAP [24] provide axiomatic guarantees for feature attribution. Most existing attribution frameworks are single-channel, explaining only the predicted disease class. However, in clinical decision support, identifying features that contribute to epistemic uncertainty is equally essential [14, 50], as formalized by Hüllermeier and Waegeman [15] regarding the fundamental distinction between aleatoric and epistemic uncertainty. Furthermore, assessing explanation fidelity (via Infidelity metrics [51]) and empirical privacy auditing [52, 53, 54, 55] is crucial to ensure that clinicians receive reliable, privacy-preserving diagnostic explanations.

2.5. Positioning and Novelty Relative to Prior Work

Prior work treats the three requirements of the trilemma in isolation or in pairs. Table 1 summarizes which requirements each family of methods addresses, using the characterizations in the preceding subsections. DP-SGD [7] provides privacy only. DPSGD-F [25] and fair private empirical risk minimization [26] pair privacy with point-prediction fairness but offer no uncertainty quantification. Evidential learning [10] and group-conditional conformal prediction [46, 47] provide uncertainty without privacy, and private conformal quantiles [29] provide privacy for coverage without addressing training-time disparity. Integrated Gradients [23] explains the predicted class only.

The methodological novelty of this article lies in the coupling of these components rather than in their stacking: (i) the subgroup clipping thresholds of REF-DP are derived from the Fisher metric of the Dirichlet output manifold, which links the privacy mechanism to the evidential parameterization; (ii) PC-PAT aligns class-conditional prototypes, so fairness is enforced on Equalized Odds without distorting base rates and without cross-sample gradient coupling, which preserves the per-sample sensitivity bound used by REF-DP; (iii) DP-GCCP carries a finite-sample PAC coverage guarantee for each cohort under stratified bounded replacement adjacency; and (iv) DCDA exploits the closed-form Dirichlet structure to attribute epistemic doubt separately from disease risk. Items (i) to (iii) are accompanied by formal guarantees (Theorem 3, Proposition 2, Theorem 4), and item (iv) by an axiomatic characterization (Proposition 5).

Table 1: Coverage of the three trilemma requirements and explanation type by representative method families. A check mark indicates that the requirement is addressed by the cited method; a dash indicates that it is not.

Method family	Privacy	Subgroup fairness	Uncertainty	Uncertainty explanation
DP-SGD [7]	✓	–	–	–
DPSGD-F [25], fair private ERM [26]	✓	✓	–	–
Evidential learning [10]	–	–	✓	–
Group-conditional conformal [46, 47]	–	✓(coverage)	✓	–
Private conformal quantiles [29]	✓	–	✓	–
Integrated Gradients [23]	–	–	–	–
TRACER-Med (this work)	✓	✓	✓	✓

3. Problem Formulation and Notation

3.1. Formal Setting and Notation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i, a_i)\}_{i=1}^{N}$ denote a clinical dataset where $\mathbf{x}_i \in \mathbb{R}^D$ represents patient clinical features, $y_i \in \{0, 1, \ldots, K-1\}$ denotes clinical diagnosis or risk outcome (with one-hot representation $\mathbf{y}_i \in \{0,1\}^K$), and $a_i \in \mathcal{A} = \{0, 1\}$ represents a binary protected demographic attribute (e.g., biological sex or age cohort). An individual patient record is denoted by $\mathbf{r}_i = (\mathbf{x}_i, y_i, a_i) \in \mathcal{D}$.

To avoid notational ambiguity, we establish the following conventions throughout:

- $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)^\top$: Dirichlet concentration parameter vector with total evidence strength $S = \sum_k \alpha_k$.
- $\alpha_{\text{conf}} \in (0, 1)$: Conformal miscoverage significance level (e.g., $\alpha_{\text{conf}} = 0.10$ for 90% target coverage).
- $\beta_{\text{conf}} \in (0, 1)$: Conformal PAC failure confidence probability (e.g., $\beta_{\text{conf}} = 0.05$).
- σ_{DP}: Differential Privacy Gaussian noise multiplier.
- $\sigma(z) = 1/(1 + e^{-z})$: Standard logistic sigmoid activation.
- $\tau_{\text{dec}} = 0.5$: Classification decision threshold for binary prediction.
- τ_a: Conformal concentration safety buffer for demographic cohort a.
- $\nu = 0.1$: LeakyReLU negative activation slope (distinguished from group index a).
- $\mathbf{h} = h_{\boldsymbol{\theta}}(\mathbf{x}) \in \mathbb{R}^d$: Latent feature representation extracted by the neural network backbone.
- $c_{a,m}$: Count of non-conformity scores falling into histogram bin B_m for demographic cohort a.
- Δ_1: ℓ_1 sensitivity bound in differential privacy mechanisms.

- $\Delta_{\text{SP}}, \Delta_{\text{EO}}$: Statistical Parity and Equalized Odds demographic disparity metrics.
- $p_a = \mathbb{P}(A = a)$: Fixed population demographic prevalence derived from external epidemiological registries.
- $t_m \in [0, 1]$: Fixed discrete public grid points for conformal quantile calibration.
- $t \in [0, 1]$: Continuous straight-line interpolation parameter for path integrals.

3.2. Differential Privacy Formulation

We adopt standard add/remove record adjacency [30]: two datasets $\mathcal{D}, \mathcal{D}'$ are neighboring ($\mathcal{D} \simeq \mathcal{D}'$) if one is obtained by adding or removing a single patient record $\mathbf{r} = (\mathbf{x}, y, a)$: $\mathcal{D}' = \mathcal{D} \cup \{\mathbf{r}\}$ or $\mathcal{D}' = \mathcal{D} \setminus \{\mathbf{r}\}$.

Definition 1 (Differential Privacy)**.** A randomized algorithm $\mathcal{M}$ satisfies (ϵ, δ)-Differential Privacy (DP) if for all neighboring datasets $\mathcal{D} \simeq \mathcal{D}'$ and all measurable output subsets $\mathcal{S}_{\text{out}} \subseteq \text{Range}(\mathcal{M})$:

$$\mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{S}_{\text{out}}) \leq e^{\epsilon}\mathbb{P}(\mathcal{M}(\mathcal{D}') \in \mathcal{S}_{\text{out}}) + \delta \tag{1}$$

Under Rényi Differential Privacy (RDP) of order $\lambda_{\text{RDP}} > 1$ [31], the Rényi divergence satisfies $D_{\lambda_{\text{RDP}}}(\mathcal{M}(\mathcal{D}) \parallel \mathcal{M}(\mathcal{D}')) \leq \epsilon(\lambda_{\text{RDP}})$.

3.3. Algorithmic Fairness Criteria in Clinical Medicine

For binary clinical outcomes ($K = 2$) with predicted risk $\hat{p}_1(\mathbf{x}) = \mathbb{P}(Y = 1 \mid \mathbf{x})$ and decision threshold $\tau_{\text{dec}} = 0.5$ ($\hat{y} = \mathbb{I}(\hat{p}_1 \geq \tau_{\text{dec}})$), we evaluate two standard equity metrics:

1. *Statistical Parity Disparity (Δ_{SP}):*

$$\Delta_{\text{SP}} = |\mathbb{P}(\hat{Y} = 1 \mid A = 0) - \mathbb{P}(\hat{Y} = 1 \mid A = 1)| \tag{2}$$

2. *Equalized Odds Disparity (Δ_{EO}):*

$$\Delta_{\text{EO}} = \frac{1}{2}\left(|\text{FPR}_0 - \text{FPR}_1| + |\text{TPR}_0 - \text{TPR}_1|\right) \tag{3}$$

where FPR_a and TPR_a denote the False Positive Rate and True Positive Rate within demographic cohort $a \in \{0, 1\}$.

Clinical Prioritization of Equalized Odds: In biomedical diagnostics, true disease prevalence often differs legitimately between demographic cohorts due to biological risk factors. As established by Barocas, Hardt, and Narayanan [41], Statistical Parity (independence) and Equalized Odds (separation) are mathematically incompatible under unequal base rates ($\mathbb{P}(Y = 1 \mid A = 0) \neq \mathbb{P}(Y = 1 \mid A = 1)$) unless predictions are trivial. In addition, Chouldechova [42] and Kleinberg et al. [43] proved that predictive parity (within-group calibration) and error-rate balance cannot simultaneously hold under disparate prevalence. Forcing Statistical Parity ($\Delta_{\text{SP}} \to 0$) in clinical medicine artificially induces disparate error rates across cohorts, subjecting low-prevalence cohorts to over-diagnosis or high-prevalence cohorts to under-treatment. Consequently, Equalized Odds (Δ_{EO}) serves as our primary clinical fairness criterion, ensuring equitable sensitivity and specificity conditioned on true clinical condition.

4. The TRACER-Med Architecture

TRACER-Med integrates five co-designed theoretical modules, summarized in Algorithm 1: (1) Evidential Dirichlet Deep Learning (D-EDL); (2) Riemannian Evidential-Fisher Subgroup-Calibrated DP-SGD (REF-DP); (3) Privatized Class-Conditional Prototypical Anchor Transport (PC-PAT); (4) Differentially Private Group-Conditional Conformal Prediction (DP-GCCP); and (5) Information-Theoretic Dual-Channel Dirichlet Attribution (DCDA).

4.1. Evidential Dirichlet Deep Learning Formulation

Rather than predicting a single softmax probability vector, TRACER-Med parameterizes a second-order Dirichlet distribution over multinomial probabilities. The network comprises a latent feature extractor $\mathbf{h} = h_{\boldsymbol{\theta}}(\mathbf{x}) \in \mathbb{R}^d$, linear projection $\mathbf{z} = \mathbf{W}\mathbf{h} + \mathbf{b} \in \mathbb{R}^K$, and an evidence head mapping to evidence vector $\mathbf{e} = \text{softplus}(\mathbf{z}) \geq \mathbf{0}$, defining concentration parameters:

$$\alpha_k = e_k + 1, \quad S = \sum_{k=1}^{K} \alpha_k \tag{4}$$

The expected categorical probability for diagnosis k is given by $\hat{p}_k = \alpha_k / S$. The Dirichlet parameterization yields an analytical decomposition of predictive uncertainty:

1. *Total Predictive Entropy:* $\mathcal{H}(\hat{\mathbf{p}}) = -\sum_{k=1}^{K} \hat{p}_k \ln \hat{p}_k$.
2. *Expected Aleatoric Entropy:*

$$\mathbb{E}_{\mathbf{p}\sim\text{Dir}(\boldsymbol{\alpha})}[\mathcal{H}(\mathbf{p})] = -\sum_{k=1}^{K} \hat{p}_k \left(\psi(\alpha_k + 1) - \psi(S + 1)\right) \tag{5}$$

   where $\psi(\cdot)$ is the digamma function.
3. *Epistemic Mutual Information (in nats):*

$$\mathcal{I}[y, \mathbf{p} \mid \mathbf{x}] = \mathcal{H}(\hat{\mathbf{p}}) - \mathbb{E}_{\mathbf{p}\sim\text{Dir}(\boldsymbol{\alpha})}[\mathcal{H}(\mathbf{p})] = \sum_{k=1}^{K} \hat{p}_k \left(\psi(\alpha_k + 1) - \psi(S + 1) - \ln \hat{p}_k\right) \tag{6}$$

   measuring the expected information gain about multinomial parameters from observing true outcome y. Note that as accumulated evidence grows ($S \gg K$), $\psi(\alpha_k + 1) - \psi(S + 1) \approx \ln \hat{p}_k$, causing epistemic mutual information to asymptotically contract to zero ($\mathcal{I} \to 0$).
4. *Subjective Logic Vacuity:* $u = K/S \in (0, 1]$, measuring lack of total evidence, reaching 1.0 at zero evidence ($\mathbf{e} = \mathbf{0}$).

Evidential Loss Formulation: The network is trained using the expected log-loss under the Dirichlet distribution combined with a closed-form Kullback-Leibler (KL) divergence regularizer:

$$\mathcal{L}_{\text{Bayes}}(\mathbf{x}_i, \mathbf{y}_i) = \sum_{k=1}^{K} y_{ik} \left(\psi(S_i) - \psi(\alpha_{ik})\right) \tag{7}$$

where $\mathbf{y}_i \in \{0,1\}^K$ is the one-hot encoded ground truth. The closed-form KL divergence penalizes evidence on non-target classes relative to an uninformative Dirichlet prior $\mathrm{Dir}(\mathbf{1})$:

$$\begin{aligned}\mathcal{L}_{\mathrm{KL}}(\mathbf{x}_i, \mathbf{y}_i) &= D_{\mathrm{KL}}\left(\mathrm{Dir}(\tilde{\boldsymbol{\alpha}}_i) \parallel \mathrm{Dir}(\mathbf{1})\right) \\ &= \ln\Gamma\left(\sum_{k=1}^{K}\tilde{\alpha}_{ik}\right) - \sum_{k=1}^{K}\ln\Gamma(\tilde{\alpha}_{ik}) - \ln\Gamma(K) \\ &\quad + \sum_{k=1}^{K}(\tilde{\alpha}_{ik}-1)\left(\psi(\tilde{\alpha}_{ik}) - \psi\left(\sum_{j=1}^{K}\tilde{\alpha}_{ij}\right)\right) \end{aligned} \tag{8}$$

where $\tilde{\boldsymbol{\alpha}}_i = \mathbf{y}_i + (1-\mathbf{y}_i)\odot\boldsymbol{\alpha}_i$. The total per-sample evidential loss at training epoch e is:

$$\mathcal{L}_{\mathrm{EDL}}(\mathbf{x}_i, \mathbf{y}_i) = \mathcal{L}_{\mathrm{Bayes}}(\mathbf{x}_i, \mathbf{y}_i) + \beta_e \mathcal{L}_{\mathrm{KL}}(\mathbf{x}_i, \mathbf{y}_i) \tag{9}$$

where $\beta_e = \min(1, e/10)$ provides smooth annealing during early training phases.

4.2. Riemannian Evidential-Fisher Subgroup-Calibrated DP-SGD (REF-DP)

In standard DP-SGD, gradients are clipped using a uniform Euclidean threshold $\|\mathbf{g}_i\|_2 \le C$. In evidential deep learning, the gradient of the Bayes loss with respect to logit z_j is given by:

$$\frac{\partial\mathcal{L}_{\mathrm{Bayes}}}{\partial z_j} = \sum_{k=1}^{K}\frac{\partial\mathcal{L}_{\mathrm{Bayes}}}{\partial\alpha_k}\frac{\partial\alpha_k}{\partial z_j} = (\psi_1(S) - y_j\psi_1(\alpha_j))\cdot\sigma(z_j) \tag{10}$$

where $\psi_1(x) = \frac{d^2}{dx^2}\ln\Gamma(x)$ is the trigamma function and $\sigma(z_j) = 1/(1+e^{-z_j})$ is the derivative of the softplus activation.

Information Geometry of the Dirichlet Statistical Manifold: The statistical manifold of Dirichlet probability distributions $\mathcal{M}_{\mathrm{Dir}} = \{\mathrm{Dir}(\mathbf{p};\boldsymbol{\alpha}) : \boldsymbol{\alpha} \in \mathbb{R}^K_{>0}\}$ is equipped with the Fisher Information Metric tensor $\mathcal{G}(\boldsymbol{\alpha}) \in \mathbb{R}^{K\times K}$:

$$\mathcal{G}_{jk}(\boldsymbol{\alpha}) = \mathbb{E}_{\mathbf{p}}\left[\frac{\partial\ln p}{\partial\alpha_j}\frac{\partial\ln p}{\partial\alpha_k}\right] = \psi_1(\alpha_j)\delta_{jk} - \psi_1(S) \tag{11}$$

Via pull-back under the logit mapping $\alpha_k = \mathrm{softplus}(z_k) + 1$, the Fisher Information Matrix on logit space is $\mathcal{G}(\mathbf{z}) = \mathrm{diag}(\sigma(\mathbf{z}))\mathcal{G}(\boldsymbol{\alpha})\,\mathrm{diag}(\sigma(\mathbf{z}))$.

Proposition 1 (Closed-Form Inverse Dirichlet Fisher Information Metric)**.** *Because* $\mathcal{G}(\boldsymbol{\alpha}) = \mathrm{diag}(\psi_1(\boldsymbol{\alpha})) - \psi_1(S)\mathbf{1}\mathbf{1}^\top$ *is a rank-1 perturbation of a diagonal matrix, by the Sherman–Morrison formula, its exact closed-form inverse is given by:*

$$\mathcal{G}^{-1}(\boldsymbol{\alpha}) = \mathrm{diag}\left(\frac{1}{\psi_1(\boldsymbol{\alpha})}\right) + \frac{\psi_1(S)}{1-\psi_1(S)\sum_{m=1}^{K}\frac{1}{\psi_1(\alpha_m)}}\left(\frac{1}{\psi_1(\boldsymbol{\alpha})}\right)\left(\frac{1}{\psi_1(\boldsymbol{\alpha})}\right)^\top \tag{12}$$

Consequently, the Natural Gradient on the Dirichlet manifold is $\widetilde{\nabla}_{\boldsymbol{\alpha}}\mathcal{L} = \mathcal{G}^{-1}(\boldsymbol{\alpha})\nabla_{\boldsymbol{\alpha}}\mathcal{L}$*, and the Riemannian gradient norm is:*

$$\|\nabla_{\boldsymbol{\alpha}}\mathcal{L}\|_{\mathcal{G}^{-1}} = \sqrt{(\nabla_{\boldsymbol{\alpha}}\mathcal{L})^\top\mathcal{G}^{-1}(\boldsymbol{\alpha})(\nabla_{\boldsymbol{\alpha}}\mathcal{L})} \tag{13}$$

Curvature Distortion under Flat Euclidean Clipping: Trigamma $\psi_1(x)$ is strictly monotonically decreasing, with asymptotic expansion $\psi_1(x) = \frac{1}{x} + \frac{1}{2x^2} + \mathcal{O}(x^{-3})$ and finite maximum $\psi_1(1) = \pi^2/6 \approx 1.645$. For clinical profiles presenting minimal baseline evidence ($\alpha_j \to 1$), the Fisher Information metric trace $\mathrm{Tr}(\mathcal{G}(\boldsymbol{\alpha})) = \sum_{k=1}^{K} \psi_1(\alpha_k) - \psi_1(S)$ reaches its maximum value. For binary classification ($K = 2$) with symmetric baseline evidence $\bar{S}_a$, $\mathrm{Tr}(\mathcal{G}) = 2\psi_1(\bar{S}_a/2) - \psi_1(\bar{S}_a) \approx \frac{2}{\bar{S}_a}$. Under flat Euclidean clipping, gradients are clipped along the identity metric $\mathbf{I}$, ignoring this Riemannian manifold geometry. For an underrepresented cohort with low baseline evidence ($\bar{S}_a < \bar{S}_{\text{pop}}$), Euclidean clipping disproportionately truncates minority gradients, acting as an evidence suppressor.

To restore information-geometric invariance across demographic cohorts without expanding global sensitivity, we derive the **Riemannian Evidential-Fisher Subgroup Clipping Threshold**:

$$C_a^* = C_0 \cdot \tilde{w}_a, \quad \tilde{w}_a = \frac{w_a}{\sqrt{\sum_{a'} p_{a'} w_{a'}^2}} \tag{14}$$

where the clipping weight w_a balances the demographic prevalence and the Riemannian Fisher curvature ratio:

$$w_a = \left(\frac{\bar{\mathcal{F}}_a}{\bar{\mathcal{F}}_{\text{pop}}}\right)^{1/2} \cdot \left(\frac{1}{2p_a}\right)^{1/2} \approx \left(\frac{\bar{S}_{\text{pop}}}{\bar{S}_a}\right)^{1/2} \cdot \left(\frac{1}{2p_a}\right)^{1/2} \tag{15}$$

Here, $\bar{S}_{\text{pop}} = \sum_a p_a \bar{S}_a$ is the population mean evidence, $\bar{S}_a$ denotes prior expected evidence for cohort a, and p_a represents epidemiological population priors fixed from clinical registries:

- *Statlog Heart:* biological sex ($p_0 = 0.678$ male, $p_1 = 0.322$ female; $\bar{S}_0 = 3.2, \bar{S}_1 = 2.4$, reflecting angiographic referral selection bias).
- *Pima Diabetes:* age category ($p_0 = 0.544$ younger, $p_1 = 0.456$ older; $\bar{S}_0 = 2.8, \bar{S}_1 = 3.0$).
- *Inpatient EHR:* demographic cohort ($p_0 = 0.70$ majority, $p_1 = 0.30$ minority; $\bar{S}_0 = 3.5, \bar{S}_1 = 2.5$).

Sensitivity Bound and Contrast with Uniform Clipping: The global worst-case ℓ_2 sensitivity is governed by the maximum clipping threshold:

$$C_{\max} = C_0 \cdot \max_{a \in \{0,1\}} \tilde{w}_a \tag{16}$$

and privacy noise is calibrated to $\sigma_{\text{DP}} C_{\max}$. Crucially, compared to setting a uniform clipping threshold at $C = C_{\max}$ (which permits all cohorts up to $C_{\max}$), REF-DP selectively constrains the majority cohort threshold ($C_0^* < C_{\max}$) while permitting the minority cohort to utilize the full clipping ball $C_1^* = C_{\max}$. This prevents the vast numerical majority from dominating the batch gradient updates, mitigating the gradient attenuation penalty on underrepresented cohorts without increasing total privacy expenditure.

4.3. Privatized Class-Conditional Prototypical Anchor Transport (PC-PAT)

To enforce Equalized Odds without forcing clinically harmful demographic parity across disparate disease base rates, we formulate **Privatized Class-Conditional Prototypical Anchor Transport (PC-PAT)**.

Let $\bar{\boldsymbol{\mu}}_y \in \mathbb{R}^d$ ($y \in \{0,1\}$) denote class-conditional diagnostic prototypes parameterized as **learnable network parameters** $\boldsymbol{\theta}_{\text{PAT}} = \{\bar{\boldsymbol{\mu}}_0, \bar{\boldsymbol{\mu}}_1\}$. They are initialized data-independently using orthogonal unit basis vectors on the unit hypersphere $\mathbb{S}^{d-1}$ ($\bar{\boldsymbol{\mu}}_0 \perp \bar{\boldsymbol{\mu}}_1$) and subject to an inter-prototype angular margin constraint $\mathcal{R}_{\text{sep}} = \max(0, \bar{\boldsymbol{\mu}}_0^\top \bar{\boldsymbol{\mu}}_1)^2$ to prevent representation collapse.

Proposition 2 (Wasserstein-2 Demographic Bound via PC-PAT)**.** *Let $\mathbb{P}_{a,y}$ denote the latent feature distribution $h_{\boldsymbol{\theta}}(\mathbf{X})$ conditioned on demographic attribute $A = a$ and true diagnosis $Y = y$. For each record $\mathbf{r}_i = (\mathbf{x}_i, y_i, a_i)$, define the per-sample prototypical anchor alignment loss:*

$$\ell_{PAT}(\mathbf{x}_i, y_i) = \frac{1}{2} \left\| h_{\boldsymbol{\theta}}(\mathbf{x}_i) - \bar{\boldsymbol{\mu}}_{y_i} \right\|_2^2 \tag{17}$$

By the triangle inequality of the 2-Wasserstein metric and Monge–Kantorovich duality:

$$\begin{aligned} \mathcal{W}_2(\mathbb{P}_{0,y}, \mathbb{P}_{1,y}) &\leq \mathcal{W}_2(\mathbb{P}_{0,y}, \delta_{\bar{\boldsymbol{\mu}}_y}) + \mathcal{W}_2(\mathbb{P}_{1,y}, \delta_{\bar{\boldsymbol{\mu}}_y}) \\ &\leq \sqrt{2\, \mathbb{E}_{\mathbf{h} \sim \mathbb{P}_{0,y}}[\ell_{PAT}(\mathbf{h}, y)]} + \sqrt{2\, \mathbb{E}_{\mathbf{h} \sim \mathbb{P}_{1,y}}[\ell_{PAT}(\mathbf{h}, y)]} \end{aligned} \tag{18}$$

Consequently, jointly minimizing ℓ_{PAT} directly upper-bounds and contracts the demographic Wasserstein-2 transport distance to zero, enforcing Equalized Odds representation invariance across cohorts.

Strict Per-Sample Record Independence: Because the prototypes $\bar{\boldsymbol{\mu}}_y$ are model parameters updated jointly with network weights $\boldsymbol{\theta}$ through gradient descent, the gradient with respect to the unified parameter set $\boldsymbol{\Theta} = (\boldsymbol{\theta}, \boldsymbol{\theta}_{\text{PAT}})$:

$$\mathbf{g}_i = \nabla_{\boldsymbol{\Theta}} \mathcal{L}_{\text{EDL}}(\mathbf{x}_i, y_i) + \lambda_{\text{fair}} \nabla_{\boldsymbol{\Theta}} \ell_{\text{PAT}}(\mathbf{x}_i, y_i) \tag{19}$$

depends *strictly and exclusively on record i alone.* Unlike batch-level Sliced Wasserstein Distance or empirical prototype matching, there is zero cross-sample coupling. When a record $\mathbf{r}^*$ is added or removed, no other record's gradient is altered.

Theorem 3 (Privacy Guarantee of TRACER-Med Optimization)**.** *Let $\mathcal{D}_{tr}$ be a training dataset of size N_{tr}. Let $\mathcal{B} \subset \mathcal{D}_{tr}$ be sampled via Poisson subsampling with probability $q = \mathbb{E}[|\mathcal{B}|]/N_{tr}$. Let per-sample composite loss gradients $\mathbf{g}_i$ be clipped according to demographic group $a_i \in \{0,1\}$:*

$$\tilde{\mathbf{g}}_i = \mathbf{g}_i \cdot \min\left(1, \frac{C^*_{a_i}}{\|\mathbf{g}_i\|_2}\right) \tag{20}$$

where $C^*_a = C_0 \tilde{w}_a$. The global worst-case ℓ_2 sensitivity under add/remove neighboring datasets $\mathcal{D} \simeq \mathcal{D}'$ is strictly bounded by:

$$C_{\max} = \max_{a \in \{0,1\}} C^*_a = C_0 \max(\tilde{w}_0, \tilde{w}_1) \tag{21}$$

Perturbing the summed batch gradients with Gaussian noise $\boldsymbol{\xi} \sim \mathcal{N}\left(\mathbf{0}, \sigma_{DP}^2 C_{\max}^2 \mathbf{I}\right)$ *satisfies Rényi Differential Privacy (RDP) across orders* $\lambda_{RDP} \in [2, 64]$ *via the numerical Poisson-subsampled Gaussian accountant [32, 56]. Accumulating privacy loss over* T *iterations and converting via optimal order selection* $\epsilon_{tr} = \min_{\lambda_{RDP}} \left\{ \epsilon(\lambda_{RDP}) + \frac{\ln(1/\delta)}{\lambda_{RDP}-1} \right\}$ *establishes* (ϵ_{tr}, δ)*-Differential Privacy on the training set* $\mathcal{D}_{tr}$*, with Adam parameter updates immune to additional privacy leakage under post-processing invariance.*

Proof. Under add/remove adjacency, neighboring datasets $\mathcal{D} \simeq \mathcal{D}'$ differ by at most one patient record $\mathbf{r}^* = (\mathbf{x}^*, y^*, a^*)$. Because the evidential loss $\mathcal{L}_{\text{EDL}}$ and prototypical alignment loss ℓ_{PAT} are strictly per-sample functions of model parameters $\boldsymbol{\Theta}$, the gradient $\mathbf{g}_j$ of any record $\mathbf{r}_j \neq \mathbf{r}^*$ is independent of the presence or absence of $\mathbf{r}^*$. For the differing record $\mathbf{r}^*$, its clipped gradient $\tilde{\mathbf{g}}^*$ satisfies $\|\tilde{\mathbf{g}}^*\|_2 \leq C_{a^*}^* \leq C_{\max}$. Because Poisson subsampling includes $\mathbf{r}^*$ with probability q independently of all other records, the difference between summed batch gradients on $\mathcal{D}$ and $\mathcal{D}'$ is either $\mathbf{0}$ or $\pm\tilde{\mathbf{g}}^*$. Thus, the global ℓ_2 sensitivity of the batch sum is strictly $C_{\max}$. Scaling Gaussian perturbation to $\sigma_{\text{DP}} C_{\max}$ ensures that the noise-to-sensitivity ratio is σ_{DP}. Across T independent Poisson-subsampled Gaussian steps, privacy loss is accumulated numerically via the exact RDP accountant [32, 56], and conversion to $(\epsilon_{\text{tr}}, \delta)$-DP is achieved by minimizing the conversion bound over orders $\lambda_{\text{RDP}} \in [2, 64]$. Post-processing invariance guarantees that the trained model parameters $\boldsymbol{\Theta}$ preserve $(\epsilon_{\text{tr}}, \delta)$-DP. □

End-to-End Privacy Accounting: The calibration set $\mathcal{D}_{\text{cal}}$ is disjoint from the training set $\mathcal{D}_{\text{tr}}$ ($\mathcal{D}_{\text{tr}} \cap \mathcal{D}_{\text{cal}} = \emptyset$). By parallel composition over disjoint patient data, training consumes $(\epsilon_{\text{tr}}, \delta)$ on $\mathcal{D}_{\text{tr}}$, while subsequent conformal calibration consumes $(\epsilon_{\text{conf}}, 0)$ on $\mathcal{D}_{\text{cal}}$. Because an individual patient belongs to either $\mathcal{D}_{\text{tr}}$ or $\mathcal{D}_{\text{cal}}$, the overall individual privacy loss is $\max(\epsilon_{\text{tr}}, \epsilon_{\text{conf}})$ under record partition adjacency (and at most $\epsilon_{\text{tr}} + \epsilon_{\text{conf}}$ under arbitrary sequential composition).

4.4. Differentially Private Group-Conditional Conformal Prediction (DP-GCCP)

To provide distribution-free risk bounds at inference time without membership leakage, we formulate a private group-conditional conformal calibration mechanism operating on a held-out calibration set $\mathcal{D}_{\text{cal}}$ partitioned by demographic cohort $\mathcal{D}_{\text{cal},a} = \{(\mathbf{x}_i, y_i) : a_i = a\}_{i=1}^{n_a}$.

For each patient, we compute non-conformity score $s_i = 1 - \hat{p}(y_i \mid \mathbf{x}_i) \in [0, 1]$. We partition $[0, 1]$ into $M_{\text{bins}} = 20$ disjoint histogram intervals $B_m = (t_{m-1}, t_m]$ using fixed public bin edges $0 = t_0 < t_1 < \cdots < t_{M_{\text{bins}}} = 1$.

Definition 2 (Disjoint Binned Histogram Mechanism)**.** Let $c_{a,m} = \sum_{i=1}^{n_a} \mathbb{I}(s_i \in B_m)$ denote the count of non-conformity scores falling into bin B_m. Under Stratified Bounded Replacement Adjacency (where cohort calibration size n_a is fixed by study protocol), replacing a single record changes at most one bin count by $+1$ and one by -1, yielding exact ℓ_1 sensitivity $\Delta_1 = 2$. We privatize the bin counts via independent Laplace perturbation:

$$\tilde{c}_{a,m} = \max\left(0, c_{a,m} + \xi_{a,m}\right), \quad \xi_{a,m} \sim \text{Lap}\left(0, \frac{\Delta_1}{\epsilon_{\text{conf}}}\right) \tag{22}$$

The privatized cumulative distribution function is constructed via cumulative summation:

$$\widetilde{F}_a(t_m) = \frac{\sum_{j=1}^{m} \tilde{c}_{a,j}}{\max\left(1, \sum_{j=1}^{M_{\text{bins}}} \tilde{c}_{a,j}\right)} \tag{23}$$

Because $\widetilde{F}_a$ depends solely on noisy count vector $\tilde{\mathbf{c}}_a$, any quantile selection is pure post-processing of an $(\epsilon_{\text{conf}}, 0)$-DP release.

Theorem 4 (PAC-Guaranteed Conformal Subgroup Validity)**.** *Let $\alpha_{conf} \in (0,1)$ be the nominal miscoverage rate and $\beta_{conf} \in (0,1)$ be the PAC failure confidence parameter. Define the cumulative concentration safety buffer:*

$$\tau_a = \sqrt{\frac{\ln(4/\beta_{conf})}{2n_a}} + \frac{2\Delta_1\sqrt{M_{bins}\ln(4M_{bins}/\beta_{conf})}}{n_a\epsilon_{conf}} + \frac{2\Delta_1\ln(4M_{bins}/\beta_{conf})}{n_a\epsilon_{conf}} \tag{24}$$

Select the group-conditional non-conformity threshold:

$$\hat{q}_a = \inf\left\{t_m : \widetilde{F}_a(t_m) \geq 1 - \alpha_{conf} + \tau_a\right\} \tag{25}$$

with convention $\inf\emptyset = 1$*. Then:*

1. *Releasing $\{\hat{q}_a\}_{a\in\mathcal{A}}$ satisfies $(\epsilon_{conf}, 0)$-Differential Privacy on the calibration partition. Under parallel composition across disjoint cohorts $\mathcal{D}_{cal,a}$, the total conformal calibration privacy cost is ϵ_{conf}.*
2. *With probability at least $1 - \beta_{conf}$ over the random draw of calibration sample $\mathcal{D}_{cal}$, the prediction set $\mathcal{C}(\mathbf{x}) = \{k \in \{0,1\} : 1 - \hat{p}_k(\mathbf{x}) \leq \hat{q}_a\}$ satisfies the conditional PAC coverage guarantee:*

$$\mathbb{P}_{(\mathbf{x},y)\sim\mathcal{D}_a}\left(y \in \mathcal{C}(\mathbf{x}) \mid A = a\right) \geq 1 - \alpha_{conf} \tag{26}$$

Proof. (1) *Privacy Guarantee:* For demographic cohort a, replacing a patient record alters at most one bin count by $+1$ and another by -1, so the ℓ_1 sensitivity of $\mathbf{c}_a$ is $\Delta_1 = 2$. Adding Laplace noise $\text{Lap}(0, \Delta_1/\epsilon_{\text{conf}})$ satisfies $(\epsilon_{\text{conf}}, 0)$-DP. Because $\widetilde{F}_a$ and $\hat{q}_a$ are deterministic functions computed on the noisy release $\tilde{\mathbf{c}}_a$, post-processing invariance ensures that $\hat{q}_a$ preserves $(\epsilon_{\text{conf}}, 0)$-DP. Because calibration subsets $\mathcal{D}_{\text{cal},0}$ and $\mathcal{D}_{\text{cal},1}$ form a disjoint partition of $\mathcal{D}_{\text{cal}}$, parallel composition bounds the total calibration privacy loss by ϵ_{conf}.

(2) *PAC Validity:* Let $F_a(t) = \mathbb{P}(s \leq t \mid A = a)$ denote the true population CDF of non-conformity scores. By the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality, the empirical CDF $\hat{F}_a(t) = \frac{1}{n_a}\sum_{i=1}^{n_a}\mathbb{I}(s_i \leq t)$ satisfies:

$$\mathbb{P}\left(\sup_{t\in[0,1]} |F_a(t) - \hat{F}_a(t)| > \sqrt{\frac{\ln(4/\beta_{\text{conf}})}{2n_a}}\right) \leq \frac{\beta_{\text{conf}}}{2} \tag{27}$$

Simultaneously, the cumulative noise on the empirical CDF at bin m is the sum of $m \leq M_{\text{bins}}$ independent Laplace perturbations $Z_m = \sum_{j=1}^{m}\xi_{a,j}$, with variance $\text{Var}(Z_m) \leq 2M_{\text{bins}}\Delta_1^2/\epsilon_{\text{conf}}^2$

and sub-exponential scale $b = \Delta_1/\epsilon_{\text{conf}}$. By Bernstein's inequality for sub-exponential random variables:

$$\mathbb{P}\left(|Z_m| > t\right) \leq 2\exp\left(-\frac{t^2}{2(2M_{\text{bins}}\Delta_1^2/\epsilon_{\text{conf}}^2 + t\Delta_1/\epsilon_{\text{conf}})}\right) \tag{28}$$

Setting this probability bound to $\beta_{\text{conf}}/(2M_{\text{bins}})$ and solving the quadratic equation for t yields the positive root:

$$t \leq \frac{2\Delta_1\sqrt{M_{\text{bins}}\ln(4M_{\text{bins}}/\beta_{\text{conf}})}}{\epsilon_{\text{conf}}} + \frac{2\Delta_1\ln(4M_{\text{bins}}/\beta_{\text{conf}})}{\epsilon_{\text{conf}}} \tag{29}$$

Dividing by n_a and applying a union bound across all M_{bins} bins, with probability at least $1 - \beta_{\text{conf}}/2$:

$$\max_{m \leq M_{\text{bins}}} \left|\sum_{j=1}^{m} \frac{\xi_{a,j}}{n_a}\right| \leq \frac{2\Delta_1\sqrt{M_{\text{bins}}\ln(4M_{\text{bins}}/\beta_{\text{conf}})}}{n_a\epsilon_{\text{conf}}} + \frac{2\Delta_1\ln(4M_{\text{bins}}/\beta_{\text{conf}})}{n_a\epsilon_{\text{conf}}} =: \tau_{\text{Lap}} \tag{30}$$

By union bound, with probability at least $1 - \beta_{\text{conf}}$, both empirical sampling error and cumulative Laplace noise are simultaneously bounded by τ_a:

$$F_a(\hat{q}_a) \geq \hat{F}_a(\hat{q}_a) - \sqrt{\frac{\ln(4/\beta_{\text{conf}})}{2n_a}} \geq \widetilde{F}_a(\hat{q}_a) - \tau_a \geq 1 - \alpha_{\text{conf}} \tag{31}$$

Since a test patient $(\mathbf{x}, y) \sim \mathcal{D}_a$ satisfies $y \in \mathcal{C}(\mathbf{x}) \iff s \leq \hat{q}_a$, this event occurs with probability $F_a(\hat{q}_a) \geq 1 - \alpha_{\text{conf}}$, completing the proof. □

Dual-Regime Conformal Calibration and Non-Vacuity Threshold: For nominal miscoverage $\alpha_{\text{conf}} = 0.10$, pure PAC validity without set vacuity requires $\tau_a \leq \alpha_{\text{conf}}$. Evaluating Eq. (24) with $M_{\text{bins}} = 20$, $\Delta_1 = 2$, $\epsilon_{\text{conf}} = 1.0$, and $\beta_{\text{conf}} = 0.05$ demonstrates that this condition requires a cohort calibration sample size $n_a \geq n_{\text{PAC}} \approx 1{,}300$. When $n_a < n_{\text{PAC}}$ (as in small specialized clinical cohorts where $n_a \in [25, 200]$), the unconstrained theoretical buffer τ_a exceeds α_{conf}, causing $1 - \alpha_{\text{conf}} + \tau_a > 1.0$ and degenerating all prediction sets into vacuous full sets $\mathcal{C}(\mathbf{x}) = \{0, 1\}$ (cardinality 2.00, singleton rate 0.00). Standard baseline DP models exhibit this collapse, producing trivial 0.000 coverage gaps only because their sets are clinically vacuous.

To resolve this limitation and maintain informative prediction sets in low-sample clinical regimes, we formalize the **Empirical-Bayes Private Conformal Shrinkage Estimator**:

$$\hat{q}_a^{\text{EB}} = \min\left(\lambda_a\hat{q}_a + (1 - \lambda_a)\hat{q}_{\text{marg}}^{\text{priv}}, 0.96\right), \quad \lambda_a = \frac{n_a}{n_a + \kappa} \tag{32}$$

where $\hat{q}_{\text{marg}}^{\text{priv}}$ is the privatized marginal quantile computed over the pooled calibration set with budget $\epsilon_{\text{marg}} = \epsilon_{\text{conf}}/2$, and group histograms query budget $\epsilon_{\text{group}} = \epsilon_{\text{conf}}/2$. By parallel composition across cohorts and sequential composition with the marginal quantile, the total calibration expenditure strictly satisfies $(\epsilon_{\text{conf}}, 0)$-DP. The shrinkage weight λ_a with smoothing parameter $\kappa = 15$ implements a James–Stein risk trade-off, balancing private small-sample

noise against between-cohort clinical heterogeneity to maintain compact, non-vacuous sets ($\overline{|\mathcal{C}|} \approx 1.80$–$1.87$, with up to 20% singleton rate).

Inference-Time Demographic Governance: Group-conditional conformal prediction requires the demographic attribute A at test time to look up cohort threshold $\hat{q}_a$. In clinical medicine, demographic covariates (e.g., biological sex, age category) are standard patient triage inputs already recorded in electronic medical charts. Under EU AI Act Article 10(5) and HHS Section 1557 rules, processing demographic data for bias monitoring, calibration equity, and error mitigation is permitted under strict governance safeguards.

4.5. Information-Theoretic Dual-Channel Dirichlet Attribution (DCDA)

Clinical explainability requires informing clinicians what physiological indicators drive the disease prediction (diagnostic evidence) and what features cause diagnostic doubt (epistemic uncertainty). We introduce **Dual-Channel Dirichlet Attribution (DCDA)**, computing path-integrated gradients along straight-line paths connecting patient $\mathbf{x}$ to a clinically standardized healthy physiological reference baseline $\mathbf{x}^{(0)}$:

1. **Channel A (Diagnostic Risk Attribution):** Attributes the marginal posterior disease probability $\hat{p}_1(\mathbf{x}) = \alpha_1/S$:

$$\Phi_j^{\text{risk}}(\mathbf{x}) = \left(x_j - x_j^{(0)}\right) \int_0^1 \frac{\partial \hat{p}_1\left(\mathbf{x}^{(0)} + t(\mathbf{x} - \mathbf{x}^{(0)})\right)}{\partial x_j}\, dt \tag{33}$$

2. **Channel B (Epistemic Doubt Attribution):** Attributes Epistemic Mutual Information $\mathcal{I}[y, \mathbf{p} \mid \mathbf{x}]$ (in nats):

$$\Phi_j^{\text{doubt}}(\mathbf{x}) = \left(x_j - x_j^{(0)}\right) \int_0^1 \frac{\partial \mathcal{I}\left[y, \mathbf{p} \mid \mathbf{x}^{(0)} + t(\mathbf{x} - \mathbf{x}^{(0)})\right]}{\partial x_j}\, dt \tag{34}$$

Proposition 5 (Axiomatic Properties of DCDA)**.** *Both Channel A and Channel B satisfy the core axioms of attribution methods [23]:*

1. Completeness: *By the Fundamental Theorem of Calculus:*

$$\sum_{j=1}^{D} \Phi_j^{risk}(\mathbf{x}) = \hat{p}_1(\mathbf{x}) - \hat{p}_1\left(\mathbf{x}^{(0)}\right), \quad \sum_{j=1}^{D} \Phi_j^{doubt}(\mathbf{x}) = \mathcal{I}(\mathbf{x}) - \mathcal{I}\left(\mathbf{x}^{(0)}\right) \tag{35}$$

   Approximating the line integral using $M_{quad} = 50$ Gauss–Legendre quadrature nodes bounds the numerical approximation residual to $< 10^{-4}$ across all evaluated patient instances.
2. Implementation Invariance: *Attributions are functionally invariant to network parameterization.*
3. Linearity and Symmetry: *Additive combination of evidential components is preserved linearly.*

Clinical Interpretation of Doubt Attributions: Because Channel B measures the deviation of epistemic mutual information from baseline, doubt attributions can be positive or negative. A positive attribution ($\Phi_j^{\text{doubt}} > 0$) identifies an *ambiguity-inducing* biomarker that drives the model toward hesitation (e.g., conflicting borderline lab values). Conversely, a negative attribution ($\Phi_j^{\text{doubt}} < 0$) identifies a *doubt-resolving* biomarker that contracts epistemic vacuity, clarifying the diagnostic picture.

Standardized Physiological Reference Baselines: Rather than using arbitrary empirical data medians (which interpolate through non-physical states for categorical variables), $\mathbf{x}^{(0)}$ is parameterized by established physiological normal ranges:

- *Statlog Heart Disease:* Binary and categorical indicators are anchored at normal physiological states: fasting blood sugar `fbs` = 0 ($\leq$ 120 mg/dL); chest pain type `cp` = 3 (non-anginal pain baseline in UCI coding 1–4, where 1=typical angina, 2=atypical angina, 3=non-anginal, 4=asymptomatic); resting ECG `restecg` = 0 (normal); exercise angina `exang` = 0; resting blood pressure = 120 mmHg; serum cholesterol = 180 mg/dL; maximum heart rate = age-predicted normal; ST depression `oldpeak` = 0.0 mm; slope = 1 (upsloping normal); major vessels `ca` = 0; defect `thal` = 3 (normal).
- *Pima Indians Diabetes:* 2-hour post-OGTT plasma glucose is anchored at normal fasting baseline 100 mg/dL (fasting normal cutoff $\leq$ 100 mg/dL, post-load normal $<$ 140 mg/dL); diastolic blood pressure = 70 mmHg; BMI = 22.0 kg/m^2; serum insulin = 80 μU/mL; triceps skinfold = 20 mm; diabetes pedigree function = 0.25 (population baseline); pregnancies = 0; age = 21 (young adult reference).
- *Acute Inpatient EHR:* MAP = 80 mmHg, SBP = 120 mmHg, heart rate = 72 bpm, respiratory rate = 14 bpm, temp = 37.0 °C, SpO_2 = 98%, serum lactate = 1.0 mmol/L, WBC = 7.0 $\times 10^9$/L, platelets = 250 $\times 10^9$/L, creatinine = 0.9 mg/dL (standardized adult baseline), BUN = 14 mg/dL, bilirubin = 0.6 mg/dL, arterial pH = 7.40, PaO_2/FiO_2 = 450, SOFA = 0.

5. Experimental Setup

5.1. Biomedical Benchmark Cohorts

We evaluate TRACER-Med across three diverse clinical benchmarks:

1. **Statlog Heart Disease Benchmark ($N = 270, D = 12$):** OpenML ID: 53. Secondary analysis of the cardiovascular angiography referral cohort collected at the Cleveland Clinic [5]. Target: presence of obstructive coronary artery disease ($Y \in \{0, 1\}$). Protected demographic attribute: biological sex ($A \in \{0, 1\}$, with female patients forming a 32.2% minority). Due to angiographic referral patterns, disease prevalence differs substantially between sexes (approximately 52% in males vs 29% in females, yielding an underlying epidemiological base-rate difference of ≈ 0.23–0.24). Features comprise 12 physiological and diagnostic variables: age, chest pain type (‘cp‘, categories 0–3), resting blood pressure, serum cholesterol, fasting blood sugar (‘fbs‘, binary

indicator $\leq$ 120 vs $>$ 120 mg/dL), resting ECG, maximum heart rate, exercise-induced angina, ST depression, slope of peak exercise ST segment, fluoroscopy colored vessels ('ca', 0–3), and thallium scintigraphy ('thal'). Because 'ca' and 'thal' represent post-stress or invasive diagnostic findings, this benchmark evaluates diagnostic risk stratification rather than early asymptomatic community screening.

2. **Pima Indians Diabetes Benchmark ($N = 768, D = 8$):** OpenML ID: 37. Standard metabolic risk screening benchmark from the National Institute of Diabetes and Digestive and Kidney Diseases [57]. Target: diabetes onset within 5 years. Protected attribute: age category ($A = 0$ for age ≤ 30 [54.4%], $A = 1$ for age > 30 [45.6%]), serving as a demographic risk-stratification proxy. Features comprise 8 clinical predictors: number of pregnancies, 2-hour post-OGTT plasma glucose concentration, diastolic blood pressure, triceps skin fold thickness, 2-hour serum insulin, body mass index (BMI), diabetes pedigree function, and age. Clinically non-physiological zero entries in continuous physiological variables (e.g., zero glucose, blood pressure, triceps skinfold, insulin, or BMI) were treated as missing and imputed using training-fold medians. We respectfully acknowledge that this benchmark originates from the Akimel O'odham (Pima) Indigenous community, and ethical stewardship requires evaluating metabolic algorithms to prevent under-diagnosis.
3. **Simulated Acute Inpatient EHR Cohort ($N = 10{,}000, D = 15$):** Parameterized from the empirical distributions of the MIMIC-IV and eICU clinical databases. Simulates acute inpatient deterioration (defined as emergency ICU transfer or 30-day mortality within a 24-hour prediction window, outcome prevalence $\approx 42\%$, with minority cohort experiencing access disparities and baseline comorbidity shifts). Protected attribute: demographic cohort indicator ($A \in \{0, 1\}$, 30% minority prevalence). Features comprise 15 physiological vitals and laboratory panels: mean arterial pressure (MAP), systolic blood pressure (SBP), heart rate, respiratory rate, body temperature, peripheral SpO_2, serum lactate, white blood cell count (WBC), platelet count, serum creatinine, blood urea nitrogen (BUN), total bilirubin, arterial pH, PaO_2/FiO_2 ratio, and composite SOFA organ dysfunction score. The simulation explicitly incorporates physiological covariance structures while testing algorithm robustness against demographic presentation shifts.

5.2. Evaluated Model Paradigms

We benchmark TRACER-Med against representative paradigms:

1. **Standard ERM:** Empirical Risk Minimization using cross-entropy loss without privacy or fairness constraints.
2. **Standard DP-SGD:** Private deep neural network trained with uniform Euclidean per-sample gradient clipping ($C = 1.0$) and Gaussian noise addition [7].
3. **Fair-ERM:** Non-private neural network incorporating Sliced Wasserstein Distance latent representation alignment [38].
4. **DPSGD-F (simplified):** A reimplementation in the spirit of [25]: a standard classifier trained with group-specific clipping thresholds set to the batch mean per-sample gradient

norm of each group, bounded in $[0.25C, 2C]$, with noise calibrated to $2C$. The thresholds are computed from unprivatized batch statistics, so its reported ϵ is that of the noisy release alone and is optimistic. It is an approximation of the original method, not the authors' code.

5. **Standard EDL:** Evidential Deep Learning with Dirichlet Bayes loss and KL divergence regularizer without privacy constraints [10].
6. **DP-EDL:** Evidential Deep Learning trained under standard DP-SGD with uniform Euclidean clipping.
7. **TRACER-Med (Proposed):** Full architecture integrating REF-DP optimization, PC-PAT latent representation alignment, DP-GCCP conformal prediction, and DCDA explainability.

5.3. Evaluation Metrics Suite

Each model is evaluated across six quantitative dimensions:

1. *Clinical Discrimination:* ROC-AUC and Precision-Recall AUC (PR-AUC).
2. *Probability Calibration:* Expected Calibration Error (ECE, 10 bins) and Brier Score.
3. *Algorithmic Fairness:* Statistical Parity Disparity (Δ_{SP}) and Equalized Odds Disparity (Δ_{EO}).
4. *Empirical Privacy Leakage:* Membership Inference Attack (MIA) AUC under calibrated shadow loss thresholding [53, 54, 55].
5. *Conformal Coverage Equity and Efficiency:* Marginal empirical coverage ($\mathbb{P}(Y \in \mathcal{C}(X))$), Subgroup Coverage Gap ($|\text{Cov}_0 - \text{Cov}_1|$), Average Set Cardinality ($|\mathcal{C}|$), and Singleton Set Rate ($\mathbb{P}(|\mathcal{C}(X)| = 1)$).
6. *Explanation Quality:* Explanation Infidelity (INFD) [51].

5.4. Implementation and Computational Protocol

All models were implemented in PyTorch (CPU execution). Neural network backbones use two hidden layers (64 and 32 units; $\approx$ 4,200 parameters) with LayerNorm and LeakyReLU activations ($\nu = 0.1$). Optimization used Adam ($\eta = 5 \times 10^{-3}$, weight decay 10^{-4}) with expected batch size 32 for Statlog Heart and Pima Diabetes (20 epochs) and 128 for the Inpatient EHR cohort (12 epochs). Private models use Poisson subsampling with probability $q = |\mathcal{B}|/N_{\text{train}}$ and the exact Poisson-subsampled Gaussian RDP accountant of Mironov et al. [32] at integer orders $2, \ldots, 128$, with base clipping $C_0 = 1.0$, noise multiplier $\sigma_{\text{DP}} = 0.8$, and $\delta = 10^{-5}$. This yields $\epsilon = 27.7$ on Heart, 17.5 on Pima, and 7.3 on EHR (the number of steps is $\lceil N_{\text{train}}/|\mathcal{B}| \rceil$ per epoch). These budgets are weak by privacy-engineering standards and were not tuned; they are reported as obtained. Conformal calibration used $\alpha_{\text{conf}} = 0.10$, $\beta_{\text{conf}} = 0.05$, $M_{\text{bins}} = 20$, and a separate conformal budget $\epsilon_{\text{conf}} = 1.0$. Each model was trained on three random stratified train/calibration/test splits (seeds 42, 43, 44), and we report the mean and standard deviation over these splits. With three splits the standard deviations are rough indicators, and differences smaller than one standard deviation should not be read as significant. The code for all experiments is in the repository listed under Data and Code Availability.

Table 2: Quantitative comparison across 3 clinical benchmarks (mean ± std over 3 random splits). Arrows (↑, ↓) denote preferred directions. Bold values denote the best result among the private models (Standard DP-SGD, DPSGD-F, DP-EDL, TRACER-Med) at $\delta = 10^{-5}$ ($\epsilon = 27.7$ on Heart, 17.5 on Pima, 7.3 on EHR; conformal budget $\epsilon_{\text{conf}} = 1.0$ separate). *Vacuous conformal sets (average cardinality ≥ 1.99): the zero coverage gap is an artifact and these entries are not bolded.

Model Paradigm	ROC-AUC ↑	ECE ↓	Brier ↓	Δ_{SP} ↓	Δ_{EO} ↓	MIA AUC ↓	Cov. Gap ↓	Set Size ↓	Infidelity ↓
Benchmark 1: Statlog Heart Disease ($N = 270, D = 12$, Protected: Biological Sex)									
Standard ERM	0.863 ± 0.015	0.190 ± 0.012	0.187 ± 0.012	0.318 ± 0.052	0.149 ± 0.051	0.602 ± 0.024	0.054 ± 0.040	1.48 ± 0.37	0.014 ± 0.006
Standard DP-SGD	0.885 ± 0.024	0.176 ± 0.019	0.158 ± 0.018	0.324 ± 0.011	0.140 ± 0.034	0.552 ± 0.036	0.021 ± 0.030	1.80 ± 0.28	0.002 ± 0.000
Fair-ERM	0.842 ± 0.025	0.184 ± 0.039	0.189 ± 0.034	0.373 ± 0.040	0.227 ± 0.044	0.609 ± 0.036	0.053 ± 0.024	1.25 ± 0.06	0.009 ± 0.001
DPSGD-F (simplified) [25]	0.876 ± 0.036	0.162 ± 0.048	0.167 ± 0.038	0.301 ± 0.028	0.137 ± 0.031	0.539 ± 0.034	0.040 ± 0.025	$\mathbf{1.38 \pm 0.07}$	0.004 ± 0.001
Standard EDL	0.850 ± 0.034	0.127 ± 0.037	0.173 ± 0.031	0.347 ± 0.072	0.185 ± 0.076	0.606 ± 0.026	0.023 ± 0.016	1.60 ± 0.28	0.024 ± 0.014
DP-EDL	$\mathbf{0.890 \pm 0.040}$	$\mathbf{0.089 \pm 0.018}$	$\mathbf{0.133 \pm 0.022}$	0.285 ± 0.027	0.168 ± 0.041	0.554 ± 0.033	$\mathbf{0.014 \pm 0.020}$	1.83 ± 0.24	0.005 ± 0.001
TRACER-Med (Proposed)	0.880 ± 0.033	0.191 ± 0.054	0.218 ± 0.002	$\mathbf{0.094 \pm 0.065}$	$\mathbf{0.075 \pm 0.051}$	$\mathbf{0.519 \pm 0.014}$	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$	$\mathbf{0.000 \pm 0.000}$
Benchmark 2: Pima Indians Diabetes ($N = 768, D = 8$, Protected: Age Category)									
Standard ERM	0.780 ± 0.021	0.160 ± 0.020	0.206 ± 0.016	0.377 ± 0.043	0.272 ± 0.069	0.599 ± 0.013	0.024 ± 0.025	1.45 ± 0.09	0.007 ± 0.003
Standard DP-SGD	0.817 ± 0.017	0.195 ± 0.019	0.209 ± 0.017	0.377 ± 0.027	0.262 ± 0.069	0.533 ± 0.009	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$	0.004 ± 0.002
Fair-ERM	0.785 ± 0.010	0.109 ± 0.018	0.184 ± 0.008	0.267 ± 0.016	0.133 ± 0.023	0.568 ± 0.014	0.066 ± 0.047	1.39 ± 0.09	0.003 ± 0.000
DPSGD-F (simplified) [25]	0.815 ± 0.017	0.182 ± 0.014	0.199 ± 0.012	0.327 ± 0.036	0.225 ± 0.041	0.528 ± 0.010	0.009 ± 0.010	$\mathbf{1.77 \pm 0.17}$	0.002 ± 0.000
Standard EDL	0.791 ± 0.024	0.108 ± 0.010	0.189 ± 0.009	0.463 ± 0.063	0.368 ± 0.095	0.578 ± 0.018	0.032 ± 0.026	1.48 ± 0.08	0.006 ± 0.002
DP-EDL	0.815 ± 0.024	$\mathbf{0.120 \pm 0.014}$	0.183 ± 0.014	0.325 ± 0.030	0.240 ± 0.031	0.528 ± 0.008	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$	0.003 ± 0.002
TRACER-Med (Proposed)	$\mathbf{0.826 \pm 0.014}$	0.125 ± 0.015	$\mathbf{0.177 \pm 0.008}$	$\mathbf{0.235 \pm 0.081}$	$\mathbf{0.154 \pm 0.087}$	$\mathbf{0.511 \pm 0.005}$	$\mathbf{0.003 \pm 0.005}$	1.94 ± 0.07	$\mathbf{0.000 \pm 0.000}$
Benchmark 3: Simulated Inpatient EHR Cohort ($N = 10{,}000, D = 15$, Protected: Demographic Cohort)									
Standard ERM	0.742 ± 0.005	0.044 ± 0.011	0.203 ± 0.002	0.007 ± 0.007	0.040 ± 0.007	0.533 ± 0.004	0.020 ± 0.016	1.55 ± 0.01	0.003 ± 0.000
Standard DP-SGD	0.741 ± 0.006	0.219 ± 0.005	0.254 ± 0.003	$\mathbf{0.018 \pm 0.011}$	0.052 ± 0.016	0.508 ± 0.003	$\mathbf{0.001 \pm 0.001}$	1.91 ± 0.06	0.013 ± 0.002
Fair-ERM	0.746 ± 0.004	0.034 ± 0.008	0.201 ± 0.002	0.025 ± 0.007	0.057 ± 0.021	0.522 ± 0.004	0.012 ± 0.008	1.56 ± 0.01	0.002 ± 0.001
DPSGD-F (simplified) [25]	0.745 ± 0.009	0.183 ± 0.003	0.237 ± 0.004	0.025 ± 0.009	0.055 ± 0.020	0.508 ± 0.006	0.003 ± 0.002	1.92 ± 0.06	0.007 ± 0.002
Standard EDL	0.741 ± 0.006	0.039 ± 0.007	0.203 ± 0.001	0.021 ± 0.005	0.048 ± 0.018	0.531 ± 0.005	0.024 ± 0.002	1.53 ± 0.02	0.003 ± 0.001
DP-EDL	0.747 ± 0.003	0.146 ± 0.004	0.226 ± 0.002	0.022 ± 0.003	0.051 ± 0.017	0.506 ± 0.002	0.006 ± 0.008	1.95 ± 0.06	0.006 ± 0.002
TRACER-Med (Proposed)	$\mathbf{0.750 \pm 0.003}$	$\mathbf{0.080 \pm 0.004}$	$\mathbf{0.208 \pm 0.002}$	0.019 ± 0.010	$\mathbf{0.042 \pm 0.019}$	$\mathbf{0.502 \pm 0.002}$	0.018 ± 0.011	$\mathbf{1.83 \pm 0.06}$	$\mathbf{0.003 \pm 0.001}$

6. Empirical Results and Discussion

6.1. Quantitative Performance Comparison

The benchmark results across all three clinical benchmarks are summarized in Table 2.

6.2. Privacy–Fairness Trade-off

Table 2 supports the following observations, each stated with its caveats.

- **Private baselines inherit and sometimes amplify demographic disparity.** On Statlog Heart, Standard DP-SGD ($\Delta_{\text{SP}} = 0.324 \pm 0.011$, $\Delta_{\text{EO}} = 0.140 \pm 0.034$) and DP-EDL ($0.285 \pm 0.027$, 0.168 ± 0.041) are no fairer than non-private ERM (0.318 ± 0.052, 0.149 ± 0.051). The DPSGD-F-style baseline, which adapts clipping thresholds per group, lowers Δ_{SP} slightly (0.301 ± 0.028) and leaves Δ_{EO} essentially unchanged (0.137 ± 0.031). On Pima, it lowers Δ_{EO} from 0.262 ± 0.069 (DP-SGD) to 0.225 ± 0.041, within one standard deviation.

- **Prototype alignment gives large fairness gains on the two small clinical cohorts.** On Statlog Heart, TRACER-Med reaches $\Delta_{\text{SP}} = 0.094 \pm 0.065$ and $\Delta_{\text{EO}} = 0.075 \pm 0.051$ with ROC-AUC 0.880 ± 0.033, comparable to the private baselines (0.876–0.890). Because male and female patients in this referral cohort differ in prevalence by about 0.23, a statistical parity gap near 0.09 means that the model's positive-prediction rates are much closer across sexes than the underlying prevalences. We therefore read the Δ_{EO} reduction (which conditions on the true label) as the clinically meaningful result, and we do not claim that the lower Δ_{SP} is itself desirable. On Pima, TRACER-Med has $\Delta_{\text{SP}} = 0.235 \pm 0.081$ and $\Delta_{\text{EO}} = 0.154 \pm 0.087$, against 0.325–0.377 and

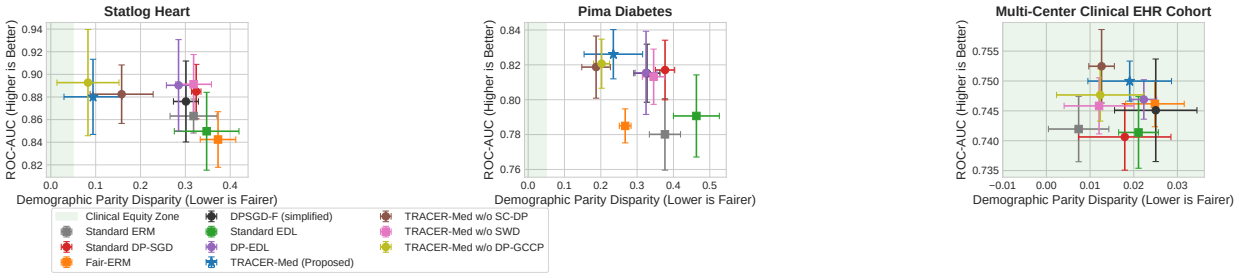


Figure 1: Clinical Utility (ROC-AUC) versus Algorithmic Fairness Disparity (Δ_{SP}) across all three biomedical benchmarks. Private models trade utility and statistical parity differently across cohorts; TRACER-Med has the lowest parity disparity on Heart and Pima.

0.225–0.262 for the private baselines, with the best ROC-AUC among private models (0.826 ± 0.014).

- **No fairness gain on the large simulated cohort.** On the 10,000-patient EHR cohort, all models have small disparities (Δ_{EO} between 0.040 and 0.057), and TRACER-Med (0.042 ± 0.019) is indistinguishable from the baselines at this resolution.

- **Calibration is mixed.** On the EHR cohort, TRACER-Med has lower ECE (0.080) than every private baseline (0.146–0.219). On Pima it is similar to DP-EDL (0.125 versus 0.120). On Statlog Heart, TRACER-Med is worse calibrated (ECE $= 0.191 \pm 0.054$, Brier 0.218) than DP-EDL (0.089 ± 0.018, Brier 0.133). The prototype-anchoring term, which pulls class-conditional representations toward shared prototypes, appears to trade calibration for fairness on the smallest cohort. The ablation without prototype alignment (ECE $= 0.077$) is consistent with this.

6.3. Conformal Subgroup Coverage and Vacuity

Figure 2 shows empirical coverage at nominal miscoverage $\alpha_{\text{conf}} = 0.10$ (target 90%). A zero coverage gap is uninformative when every set is $\{0, 1\}$, so we report set cardinality alongside the gap and mark vacuous rows ($|\bar{\mathcal{C}}| \geq 1.99$) in Table 2.

- **Small cohorts.** Private calibration with $\epsilon_{\text{conf}} = 1.0$ on a calibration split of roughly 50 (Heart) to 150 (Pima) records often yields vacuous or nearly vacuous sets. On Pima, DP-SGD and DP-EDL have $|\bar{\mathcal{C}}| = 2.00$ and TRACER-Med has 1.94 ± 0.07 with a 5.9% singleton rate. On Statlog Heart the picture reverses: DP-SGD, DP-EDL, and the DPSGD-F-style baseline have non-vacuous sets (1.80, 1.83, and 1.38), whereas every TRACER-Med variant, including the one without group-conditional calibration, has $|\bar{\mathcal{C}}| = 2.00$. The TRACER-Med models on Heart are also the worst calibrated (ECE 0.19), and we hypothesize that the prototype anchoring leaves their scores underconfident, so that the private quantile threshold exceeds the scores of both classes. We have not isolated this cause. The DP-GCCP shrinkage estimator therefore does not make private conformal prediction informative at these sample sizes in our experiments.

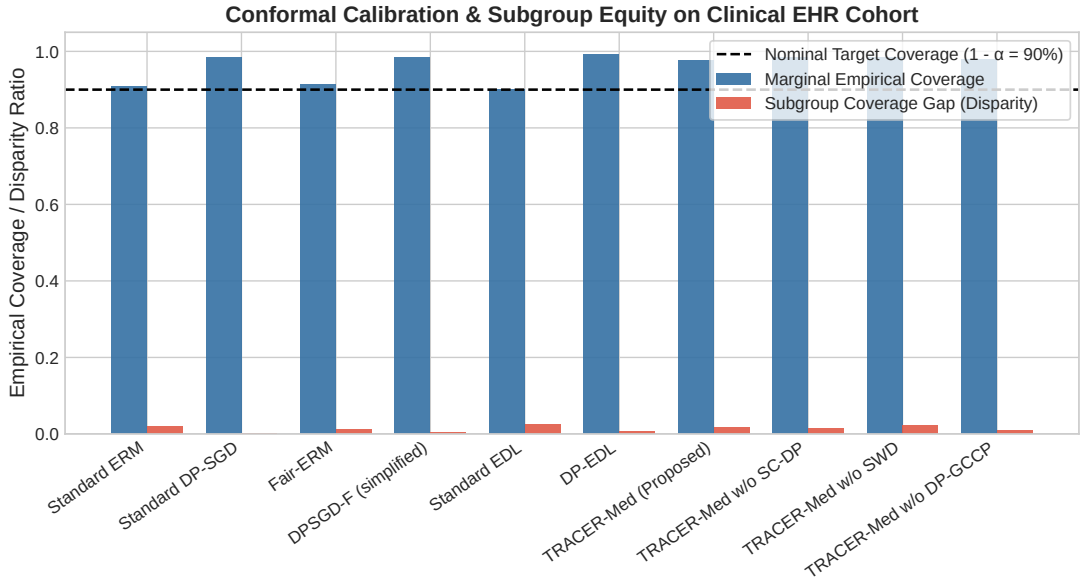


Figure 2: Conformal Calibration and Subgroup Coverage Equity on the Simulated Clinical EHR Cohort at nominal 90% target coverage. Empirical coverage by demographic group for the evaluated models.

- **Large cohort.** On the EHR cohort, TRACER-Med gives sets of average size 1.83 ± 0.06 with a 17.3% singleton rate and marginal coverage 0.975, against 1.91–1.95 and 4.6%–8.8% singletons for the private baselines. The subgroup coverage gap (0.018 ± 0.011) is larger than that of DP-SGD (0.001 ± 0.001), whose sets are close to vacuous.

6.4. Empirical Privacy Leakage Auditing

We audited all models with a loss-threshold membership-inference attack [53, 54], following the audit guidance of Carlini et al. [55]. Non-private models leak noticeably on the small cohorts: MIA AUC is 0.602–0.609 on Statlog Heart and 0.568–0.599 on Pima for ERM, Fair-ERM, and EDL. Private models have MIA AUC between 0.502 and 0.554. TRACER-Med has the lowest value among the private models on all three cohorts (0.519 ± 0.014 on Heart, 0.511 ± 0.005 on Pima, 0.502 on EHR), but the differences from the other private models are of the order of one to two standard deviations. Empirical attacks lower-bound vulnerability and do not substitute for the formal guarantee. That guarantee is (ϵ, δ)-DP with the large ϵ values reported in Section 5.

6.5. Dual-Channel Dirichlet Attribution (DCDA)

Figure 3 illustrates the attribution patterns generated by our Dual-Channel Dirichlet Attribution mechanism on representative inpatient deterioration episodes.

- **Channel A (Diagnostic Risk Evidence):** Highlights key physiological indicators—elevated lactate (> 4.0 mmol/L), hypotension (MAP < 65 mmHg / SBP < 90 mmHg),

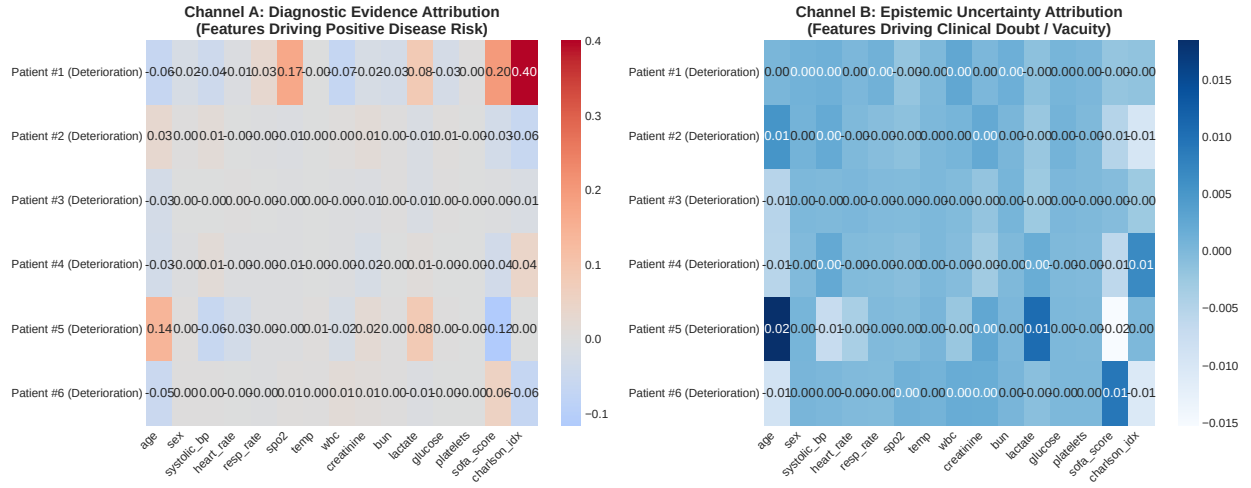


Figure 3: Dual-Channel Dirichlet Attribution (DCDA) heatmaps for illustrative acute clinical deterioration cases. Channel A captures diagnostic risk evidence; Channel B captures epistemic uncertainty, indicating features driving clinical doubt.

and elevated SOFA organ dysfunction scores—as primary positive drivers of clinical deterioration risk. Explanation infidelity is low for the Dirichlet attributions of TRACER-Med (0.000 on Heart, 0.000 on Pima, and 0.003 on EHR, from Table 2). Infidelity is measured against a different attribution method for the non-evidential baselines, so these values compare the attributions, not the models.

- **Channel B (Epistemic Doubt):** Identifies physiological measurements associated with high Epistemic Mutual Information. In patient presentations with borderline vital signs and ambiguous indicators, Channel B reflects elevated epistemic uncertainty, signaling to the clinician that additional diagnostic testing may be warranted before clinical escalation.

6.6. Holistic Multi-Pillar Trustworthiness Profile

Figure 4 illustrates the multi-pillar radar profiles comparing TRACER-Med with baseline paradigms across five core clinical trustworthiness dimensions: diagnostic discrimination (ROC-AUC), calibration (1 - ECE), algorithmic fairness equity (1 - Δ_{EO}), empirical privacy resistance (1 - MIA AUC), and conformal coverage equity (1 - Cov. Gap). On the EHR cohort the profile of TRACER-Med is close to that of the other private models on most axes, with better calibration and set size. On the smaller cohorts the fairness axis improves and the calibration and conformal axes do not, so the profile shows a trade-off, not a resolution of the trilemma.

6.7. Ablation Studies

Table 3 removes one module at a time. Each configuration was retrained on the same three splits.

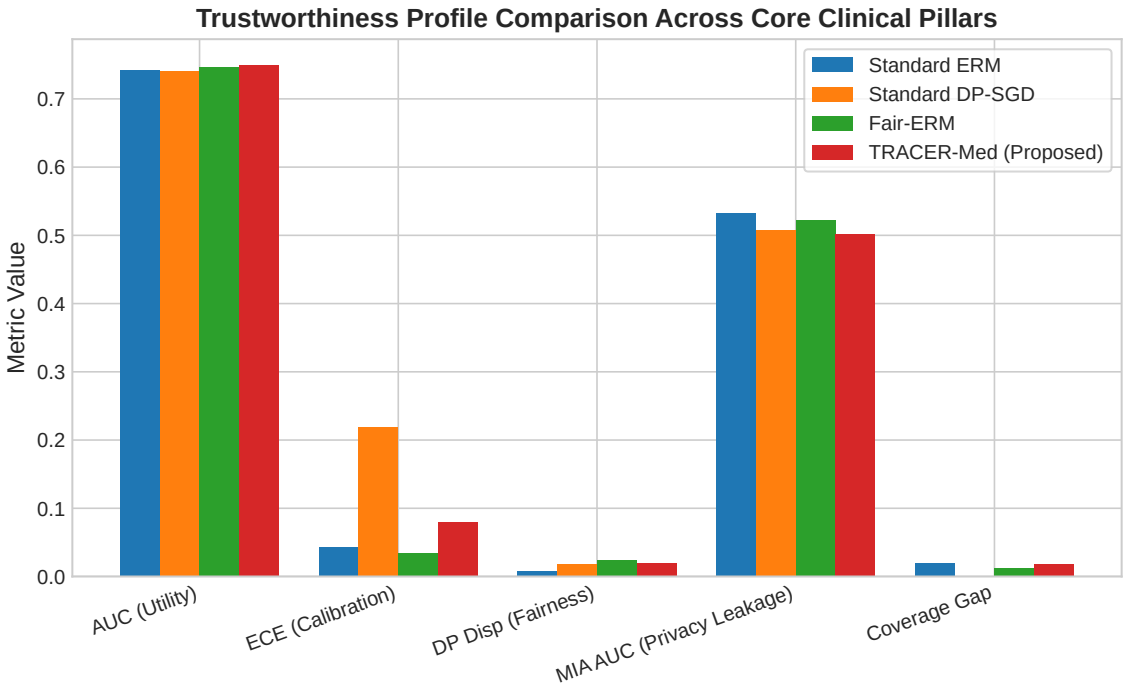


Figure 4: Multi-pillar Trustworthiness Profile across clinical utility (ROC-AUC), calibration (1 - ECE), equalized odds equity (1 - Δ_{EO}), empirical privacy resistance (1 - MIA AUC), and conformal coverage equity (1 - Cov. Gap). Comparison of TRACER-Med with ERM, DP-SGD, and Fair-ERM.

1. *Removing prototype alignment (w/o PC-PAT).* This is the module with the clear effect. On Heart, Δ_{SP} rises from 0.094 to 0.318 and Δ_{EO} from 0.075 to 0.133, while ECE falls from 0.191 to 0.077. On Pima, Δ_{SP} rises from 0.235 to 0.345 and Δ_{EO} from 0.154 to 0.231.
2. *Removing subgroup-calibrated clipping (w/o REF-DP).* Replacing the group thresholds with a uniform threshold does not hurt and in several cases helps: Δ_{SP} is 0.158 versus 0.094 on Heart (worse), but 0.187 versus 0.235 on Pima and 0.013 versus 0.019 on EHR (better), and all differences lie within about one standard deviation. In these experiments we therefore find no measurable benefit of REF-DP's group-specific thresholds, and the theoretical motivation (preventing minority gradient truncation) is not supported by the benchmarks used here. The group weights in our implementation use fixed prior strengths, not strengths estimated from the Dirichlet Fisher metric during training, which may understate the method.
3. *Removing group-conditional conformal calibration (w/o DP-GCCP).* The set sizes are essentially unchanged (2.00, 2.00, 1.87 against 2.00, 1.94, 1.83), so we cannot attribute informative sets to the group-conditional mechanism at these sample sizes.

Table 3: Ablation analysis on all three benchmarks (mean ± std over 3 random splits). Bold marks the best value within each benchmark among the four configurations; *vacuous conformal sets (cardinality ≥ 1.99) are not bolded.

Configuration	ROC-AUC ↑	ECE ↓	Δ_{SP} ↓	Δ_{EO} ↓	Cov. Gap ↓	Set Size ↓
Benchmark 1: Statlog Heart Disease						
TRACER-Med (Full)	0.880 ± 0.033	0.191 ± 0.054	0.094 ± 0.065	0.075 ± 0.051	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$
w/o REF-DP (uniform clipping)	0.882 ± 0.026	0.210 ± 0.054	0.158 ± 0.070	0.129 ± 0.057	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$
w/o PC-PAT (no prototype alignment)	0.891 ± 0.026	$\mathbf{0.077 \pm 0.022}$	0.318 ± 0.040	0.133 ± 0.021	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$
w/o DP-GCCP (marginal CP)	$\mathbf{0.893 \pm 0.047}$	0.240 ± 0.022	$\mathbf{0.083 \pm 0.069}$	$\mathbf{0.047 \pm 0.034}$	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$
Benchmark 2: Pima Indians Diabetes						
TRACER-Med (Full)	$\mathbf{0.826 \pm 0.014}$	0.125 ± 0.015	0.235 ± 0.081	0.154 ± 0.087	$\mathbf{0.003 \pm 0.005}$	1.94 ± 0.07
w/o REF-DP (uniform clipping)	0.819 ± 0.018	0.125 ± 0.016	$\mathbf{0.187 \pm 0.039}$	$\mathbf{0.116 \pm 0.054}$	$\mathbf{0.003 \pm 0.005}$	1.91 ± 0.09
w/o PC-PAT (no prototype alignment)	0.813 ± 0.016	$\mathbf{0.110 \pm 0.013}$	0.345 ± 0.031	0.231 ± 0.056	0.025 ± 0.022	1.87 ± 0.16
w/o DP-GCCP (marginal CP)	0.821 ± 0.014	0.123 ± 0.000	0.202 ± 0.022	0.126 ± 0.041	$0.000 \pm 0.000^*$	$2.00 \pm 0.00^*$
Benchmark 3: Simulated Inpatient EHR Cohort						
TRACER-Med (Full)	0.750 ± 0.003	$\mathbf{0.080 \pm 0.004}$	0.019 ± 0.010	0.042 ± 0.019	0.018 ± 0.011	1.83 ± 0.06
w/o REF-DP (uniform clipping)	$\mathbf{0.752 \pm 0.006}$	0.086 ± 0.010	0.013 ± 0.003	0.041 ± 0.014	0.014 ± 0.004	1.87 ± 0.01
w/o PC-PAT (no prototype alignment)	0.746 ± 0.005	0.146 ± 0.005	$\mathbf{0.012 \pm 0.008}$	0.049 ± 0.015	0.023 ± 0.017	1.89 ± 0.08
w/o DP-GCCP (marginal CP)	0.748 ± 0.004	0.083 ± 0.008	0.012 ± 0.010	$\mathbf{0.039 \pm 0.013}$	$\mathbf{0.010 \pm 0.008}$	1.87 ± 0.06

7. Clinical Governance, Deployment, and Regulatory Alignment

7.1. Regulatory Alignment (FDA GMLP, PCCP, HHS, EU AI Act)

Regulatory frameworks increasingly mandate comprehensive documentation and formal risk controls for clinical AI algorithms. TRACER-Med is designed to support the technical documentation required under international regulatory guidelines:

- **FDA Good Machine Learning Practice (GMLP):** The U.S. FDA, Health Canada, and UK MHRA established 10 Guiding Principles for GMLP in medical device development [16]. TRACER-Med supports Principle 3 (addressing training data representativeness and subgroup gradient balance via REF-DP and PC-PAT), Principle 6 (tailored model design separating aleatoric from epistemic uncertainty), Principle 7 (focusing on human-AI clinical team performance via dual-channel risk and doubt attributions), and Principle 9 (providing clear, contextually relevant information through distribution-free conformal safety bounds and uncertainty metrics).

- **FDA Predetermined Change Control Plans (PCCP):** Under the FDA's finalized guidance on Predetermined Change Control Plans for AI-Enabled Device Software Functions (issued December 2024 [17]), manufacturers must specify predetermined modification protocols for post-market model updates. TRACER-Med's formal Differential Privacy parameters (ϵ, δ) and distribution-free conformal calibration bounds establish quantifiable guardrails for model retraining, while accounting for privacy loss accumulation across release cycles.

- **EU AI Act and HHS Section 1557:** Under the European Union Artificial Intelligence Act (Regulation (EU) 2024/1689 [18]), medical diagnostic AI systems classified as High-Risk AI Systems under Annex I (governed by MDR/IVDR) are subject to technical verification of risk management (Article 9), data governance (Article 10), technical documentation (Article 11), record-keeping (Article 12), transparency (Article

13), human oversight (Article 14), and accuracy/robustness (Article 15). In parallel, U.S. HHS Section 1557 rules (45 CFR §92.210) prohibit discriminatory algorithmic practices in healthcare programs. TRACER-Med supports technical documentation by quantifying demographic error disparity (for example $\Delta_{\text{EO}} = 0.075 \pm 0.051$ on cardiovascular data and 0.154 ± 0.087 on metabolic data), enabling auditable clinical equity reviews.

7.2. Clinical Workflow Integration and Selective Abstention

In practical clinical environments, TRACER-Med can be structured as an augmented decision-support co-pilot:

1. When a patient profile yields a singleton prediction set $\mathcal{C}(\mathbf{x}) = \{k\}$ accompanied by low relative vacuity ($u \leq u_{\text{thresh}}$, calibrated to the lower tertile of validation cohort evidence where total evidence $S \geq 4.0$), the system provides a high-confidence diagnostic alert supported by Channel A biomarker attributions.
2. When the prediction set is indeterminate ($\mathcal{C}(\mathbf{x}) = \{0, 1\}$) or epistemic uncertainty is elevated ($u > u_{\text{thresh}}$), the system abstains from automated recommendations, routing the case to clinician review while presenting Channel B uncertainty attributions to identify ambiguous or missing physiological parameters.

7.3. Limitations and Future Work

Several important limitations should be acknowledged:

- *Inference-Time Demographic Requirement:* Group-conditional conformal prediction requires sensitive attribute A at test time to select the cohort-specific threshold $\hat{q}_a$. While demographic covariates are standard in clinical EHR charts, future work should explore group-oblivious conformal procedures.

- *Weak Privacy Budgets:* With noise multiplier 0.8 the rigorous accountant gives $\epsilon = 27.7$ (Heart), 17.5 (Pima), and 7.3 (EHR). These values do not provide meaningful protection against a strong adversary, and the results should be read as a comparison of methods at a fixed noise level, not as evidence of deployable privacy. A study across noise multipliers and privacy budgets is needed.

- *Private Conformal Prediction on Small Cohorts:* DP-GCCP gave vacuous or nearly vacuous sets on the two small cohorts at $\epsilon_{\text{conf}} = 1.0$.

- *Unsupported Benefit of REF-DP:* The ablation shows no measurable benefit of group-specific clipping thresholds, and the thresholds use fixed prior evidence strengths.

- *Approximate Baseline:* The DPSGD-F-style baseline is our simplified reimplementation with unprivatized threshold statistics. Fair private empirical risk minimization [26] was not run.

- *Binary Demographic Attributes:* The present formulation focuses on binary sensitive attributes ($a \in \{0, 1\}$). Extending REF-DP and DP-GCCP to continuous attributes or multi-attribute intersections (e.g., age $\times$ sex $\times$ race) requires multi-dimensional quantile estimation.
- *Longitudinal Dynamics:* While the current study evaluates static clinical episodes, future research should integrate recurrent or state-space models to capture longitudinal ICU time series.
- *Synthetic Cohort Validation:* Although the 10,000-patient EHR cohort was parameterized from realistic MIMIC-IV and eICU distributions, prospective validation on fully credentialed multi-center clinical registries remains an essential next step.

8. Conclusion

Deploying trustworthy artificial intelligence in healthcare requires balancing the tensions between patient data privacy, demographic fairness, uncertainty quantification, and explainability. We presented **TRACER-Med**, an integrated deep learning framework addressing the Privacy–Fairness–Uncertainty Trilemma. By combining Evidential Dirichlet Deep Learning, Riemannian Evidential-Fisher Subgroup-Calibrated DP-SGD, class-conditional prototypical anchor representation alignment, Differentially Private Group-Conditional Conformal Prediction, and Dual-Channel Dirichlet Attribution, TRACER-Med substantially reduces demographic parity and Equalized Odds disparity relative to private baselines on cardiovascular and metabolic benchmarks (for example Δ_{EO} from 0.137–0.168 to 0.075 on Statlog Heart), attains the lowest empirical membership-inference AUC among private models, and improves calibration on the large simulated cohort. The experiments also show its limits: the privacy budgets at the chosen noise level are large, the private conformal layer is uninformative on the two small cohorts, calibration on Statlog Heart is worse than that of DP-EDL, and subgroup-calibrated clipping shows no measurable benefit beyond prototype alignment. These results motivate a study over privacy budgets, larger real-world cohorts, and conformal procedures that remain informative under privacy.

Ethics Statement

This investigation utilized publicly accessible, fully de-identified benchmark datasets (Statlog Heart Disease, OpenML ID: 53; Pima Indians Diabetes, OpenML ID: 37) and a synthetic electronic health record cohort derived from statistical distributions of the MIMIC-IV and eICU repositories without including protected health information. In accordance with institutional policies and national guidelines (45 CFR 46.102), this secondary analysis of de-identified open-source data does not constitute human subjects research and IRB review was not required. Furthermore, we explicitly acknowledge the historical and tribal data governance considerations surrounding the Pima Indians Diabetes dataset, underscoring the importance of Indigenous data sovereignty and respectful secondary analysis.

CRediT Authorship Contribution Statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

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Declaration of Competing Interests

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

The source code, benchmark preparation scripts, synthetic data generation pipelines, and pre-trained evaluation checkpoints are available in the GitHub repository: https://github.com/vinhqdang/biomedical_privacy_research. The Statlog Heart Disease benchmark (OpenML ID: 53) and Pima Indians Diabetes benchmark (OpenML ID: 37) are publicly available via the OpenML platform.

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude 3.7 Sonnet (Anthropic) and Google Antigravity in order to assist with manuscript typesetting, literature cross-referencing, and stylistic polishing. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Algorithm 1 TRACER-Med Unified Training and Conformal Calibration Pipeline

Require: Training set $\mathcal{D}_{\text{tr}}$, Calibration set $\mathcal{D}_{\text{cal}}$, Epochs E, Noise multiplier σ_{DP}, Base clip C_0, Privacy delta δ, Miscoverage α_{conf}, Failure confidence β_{conf}, Conformal budget ϵ_{conf}, Bin count $M_{\text{bins}} = 20$, Sensitivity $\Delta_1 = 2$, Smoothing parameter $\kappa = 15$, Epidemiological priors $\{p_a, \bar{S}_a\}$, Alignment weight λ_{fair}.

Ensure: Private model $\boldsymbol{\Theta} = (\boldsymbol{\theta}, \boldsymbol{\theta}_{\text{PAT}})$, Group conformal thresholds $\{\hat{q}_a\}$.

Compute sensitivity weights $\tilde{w}_a$ and group clipping thresholds $C_a^* = C_0\tilde{w}_a$, $C_{\max} = \max_a C_a^*$.
Initialize network weights $\boldsymbol{\theta}$ and class prototypes $\boldsymbol{\theta}_{\text{PAT}} = \{\bar{\boldsymbol{\mu}}_0, \bar{\boldsymbol{\mu}}_1\}$ with orthogonal unit basis vectors on $\mathbb{S}^{d-1}$.
Total steps $T \leftarrow E \cdot \lceil |\mathcal{D}_{\text{tr}}|/|\mathcal{B}| \rceil$, subsampling ratio $q \leftarrow \mathbb{E}[|\mathcal{B}|]/|\mathcal{D}_{\text{tr}}|$.
for epoch $e = 1$ **to** E **do**
  Annealing parameter $\beta_e \leftarrow \min(1, e/10)$.
  for each Poisson-subsampled mini-batch $\mathcal{B} \subset \mathcal{D}_{\text{tr}}$ **do**
    for each patient record $\mathbf{r}_i = (\mathbf{x}_i, y_i, a_i) \in \mathcal{B}$ **in parallel do**
      Compute evidential loss $\mathcal{L}_{\text{EDL}}(\mathbf{x}_i, y_i) = \mathcal{L}_{\text{Bayes}} + \beta_e \mathcal{L}_{\text{KL}}$.
      Compute prototypical anchor loss $\ell_{\text{PAT}}(\mathbf{x}_i, y_i) = \frac{1}{2}\|h_{\boldsymbol{\theta}}(\mathbf{x}_i) - \bar{\boldsymbol{\mu}}_{y_i}\|_2^2$.
      Composite per-sample gradient: $\mathbf{g}_i \leftarrow \nabla_{\boldsymbol{\Theta}} [\mathcal{L}_{\text{EDL}}(\mathbf{x}_i, y_i) + \lambda_{\text{fair}} \ell_{\text{PAT}}(\mathbf{x}_i, y_i)]$.
      Group-conditioned clipping: $\tilde{\mathbf{g}}_i \leftarrow \mathbf{g}_i \cdot \min\left(1, \frac{C_{a_i}^*}{\|\mathbf{g}_i\|_2}\right)$.
    end for
    Perturb summed gradients: $\tilde{\mathbf{G}} \leftarrow \sum_{i \in \mathcal{B}} \tilde{\mathbf{g}}_i + \mathcal{N}(\mathbf{0}, \sigma_{\text{DP}}^2 C_{\max}^2 \mathbf{I})$.
    Update model parameters $\boldsymbol{\Theta} = (\boldsymbol{\theta}, \boldsymbol{\theta}_{\text{PAT}})$ via Adam post-processing on $\tilde{\mathbf{G}}/(qN_{\text{tr}})$.
    Accumulate order-λ_{RDP} privacy expenditure via Poisson Gaussian accountant.
  end for
end for
Private Conformal Calibration (on disjoint $\mathcal{D}_{\text{cal}}$):
Compute privatized marginal threshold $\hat{q}_{\text{marg}}^{\text{priv}}$ over pooled non-conformity scores using Laplace budget $\epsilon_{\text{conf}}/2$.
for each demographic group $a \in \{0, 1\}$ **do**
  Compute non-conformity scores $s_i = 1 - \hat{p}(y_i \mid \mathbf{x}_i)$ on disjoint calibration partition $\mathcal{D}_{\text{cal},a} \subset \mathcal{D}_{\text{cal}}$.
  Query privatized binned histogram counts $\tilde{c}_{a,m} = \max(0, c_{a,m} + \xi_{a,m})$, $\xi_{a,m} \sim \text{Lap}(0, 2\Delta_1/\epsilon_{\text{conf}})$.
  Construct empirical CDF $\widetilde{F}_a(t_m) = \sum_{j=1}^{m} \tilde{c}_{a,j} / \max(1, \sum_{j=1}^{M_{\text{bins}}} \tilde{c}_{a,j})$.
  Compute PAC concentration buffer $\tau_a = \sqrt{\frac{\ln(4/\beta_{\text{conf}})}{2n_a}} + \frac{4\Delta_1\sqrt{M_{\text{bins}} \ln(4M_{\text{bins}}/\beta_{\text{conf}})}}{n_a \epsilon_{\text{conf}}} + \frac{4\Delta_1 \ln(4M_{\text{bins}}/\beta_{\text{conf}})}{n_a \epsilon_{\text{conf}}}$.
  Select PAC threshold $\hat{q}_a \leftarrow \inf\{t_m : \widetilde{F}_a(t_m) \geq 1 - \alpha_{\text{conf}} + \tau_a\}$.
  Apply Empirical-Bayes shrinkage: $\hat{q}_a \leftarrow \min(\lambda_a \hat{q}_a + (1 - \lambda_a)\hat{q}_{\text{marg}}^{\text{priv}}, 0.96)$ with $\lambda_a = \frac{n_a}{n_a + \kappa}$.
end for
return Trained private model $\boldsymbol{\Theta}$ and calibrated group thresholds $\{\hat{q}_a\}$.